Image Generation with Vision Language Models

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- Diffusion : generate images from random noise
- Flow Models
- Multimodal guided generation
- CLIP: Contrastive Language-Image Pretraining
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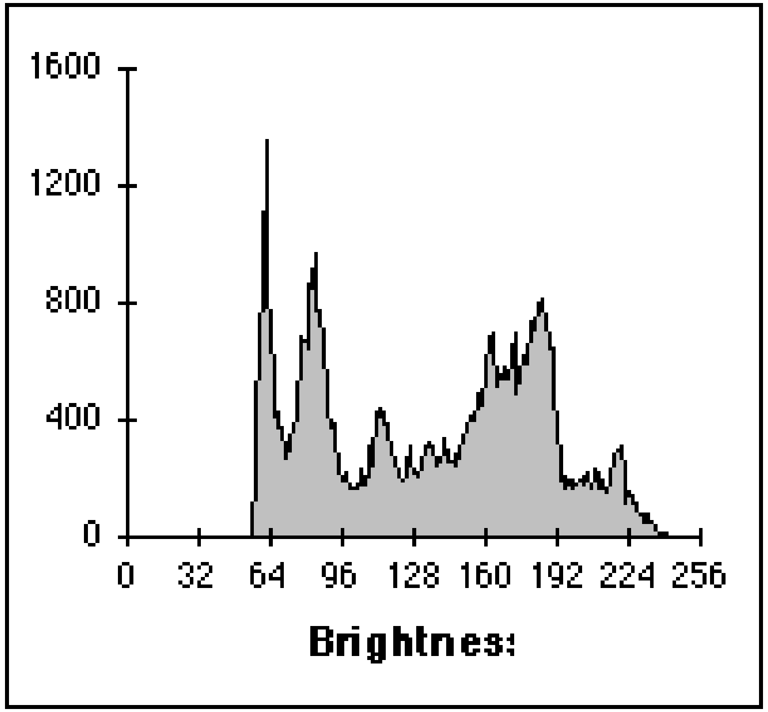
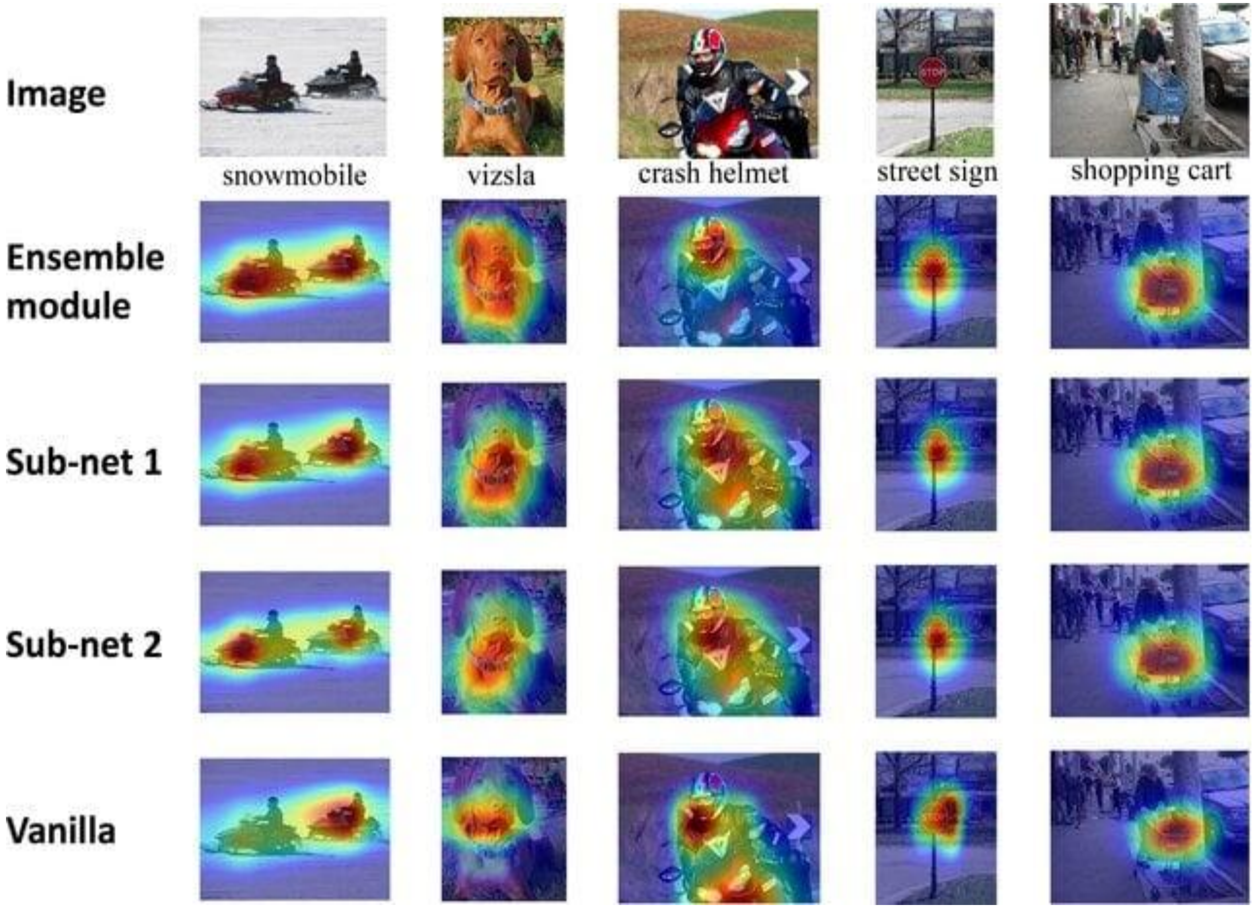
- VLM Evaluation
- Open Challenges in VLMS

Image Data Distribution

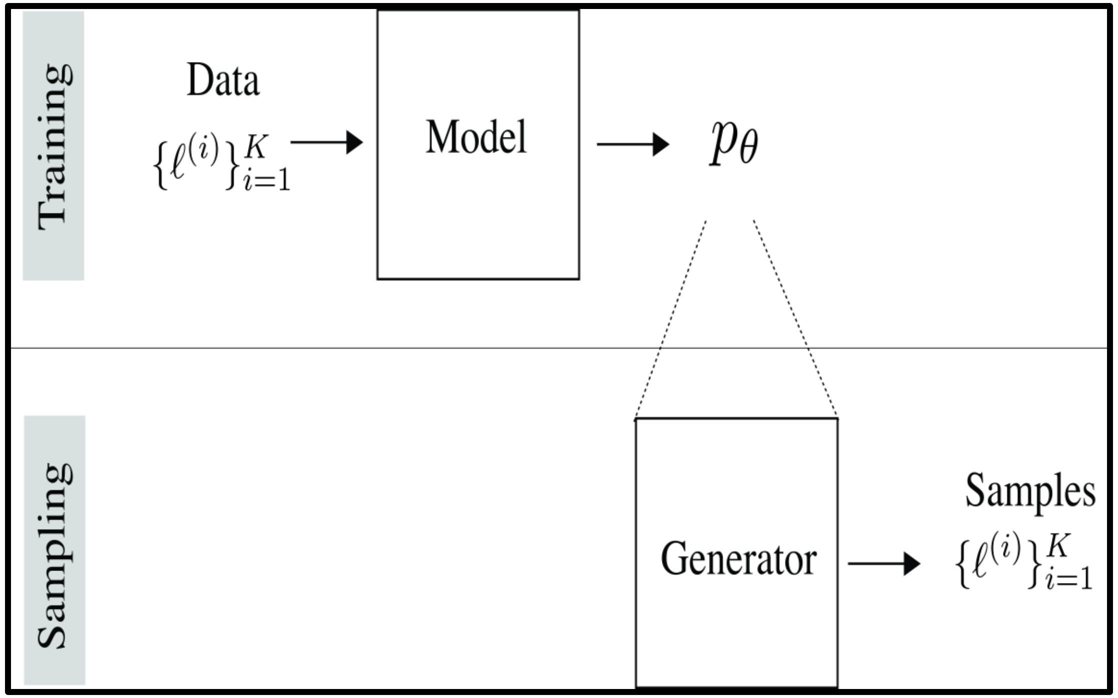


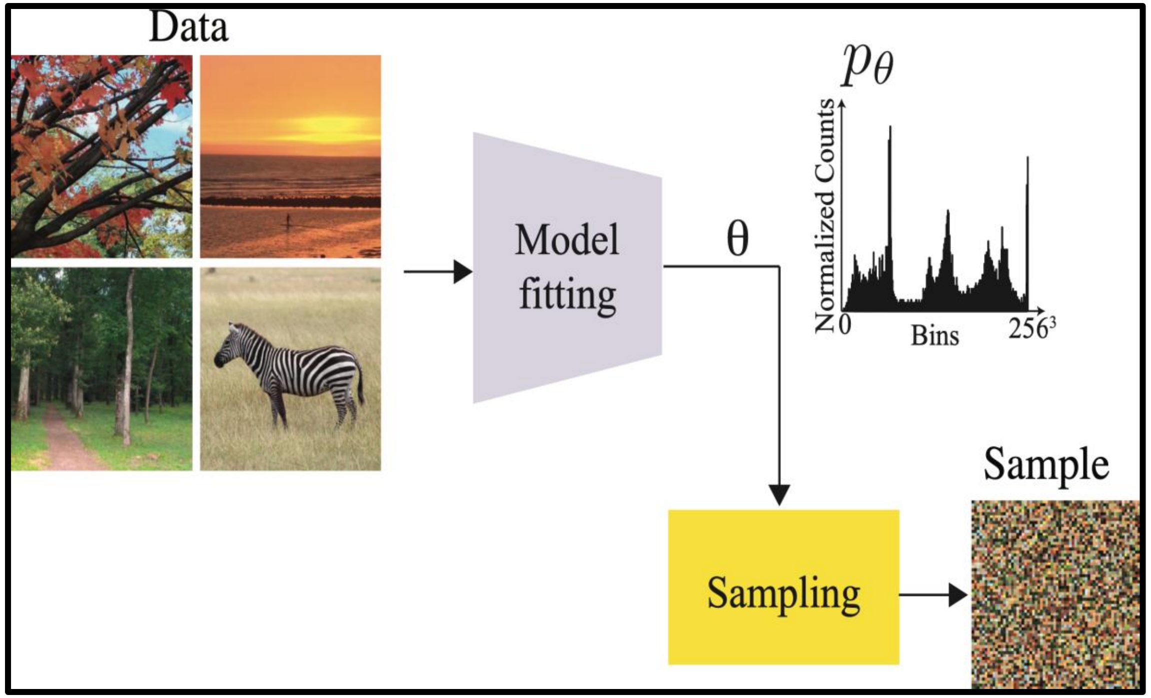


Comparison of Grad-CAM visualizations for the ensemble module and the sub-networks with the vanilla network on the ImageNet dataset

Source Credits: https://www.researchgate.net/publication/369012612_Improving_knowledge_distillation_via_pseudo-multi-teacher_network/figures?lo=1

Image Generation with Generative AI Models





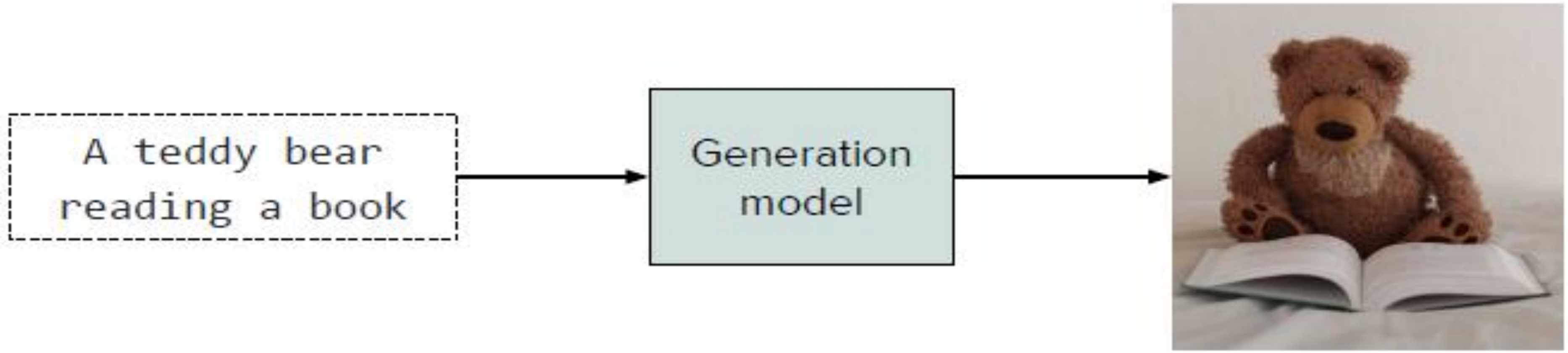
Estimation of the image model parameters using one image, and then sampling a new image by sampling pixels independently using the histogram of the training image.

Source Credits: https://visionbook.mit.edu/stat_image_models_revised.html

Vision Language Models (VLMs)

- Multimodal AI models trained to simultaneously process, align, and generate information across both visual and text domains.

- They convert text prompts into rich latent representations that guide image synthesis engines (such as diffusion or autoregressive models) to generate corresponding visual output.

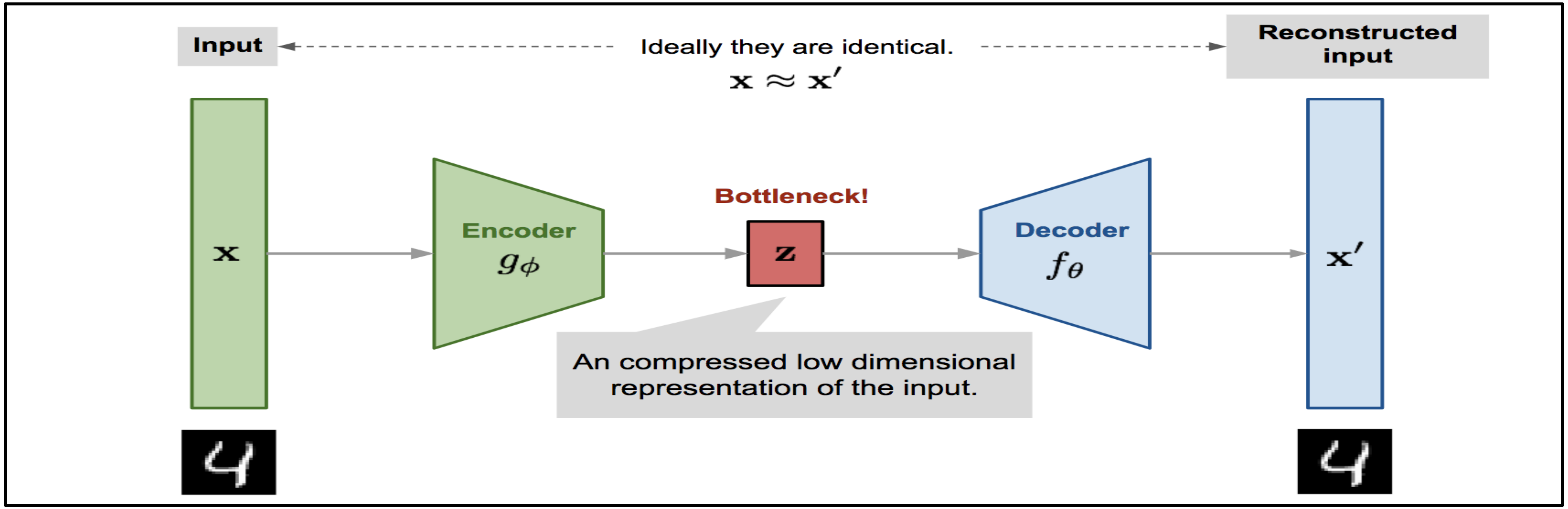


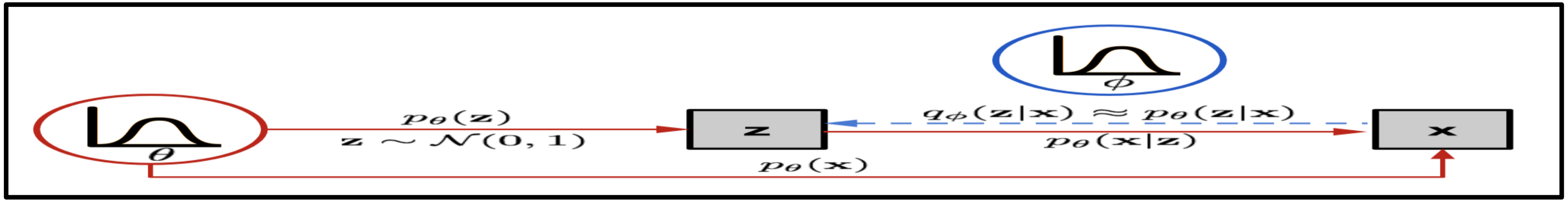
Source Credits: https://cme296.stanford.edu/

Prominent Image Generation Paradigms

- Variational Autoencoders (VAEs)
- Generative Adversarial Networks (GANs), StyleGAN (v1–v3), BigGAN.
- Diffusion Models (DDPM / Score-Based Models), Stable Diffusion v1.5 / v2.1, Midjourney (v4–v5), Imagen.
- Continuous Flow Matching (Rectified Flow / CNFs), **FLUX.1**, **Stable Diffusion 3**, SDXL Turbo
- Autoregressive Visual Models (Visual AR / Next-Token Prediction), **LlamaGen**, **VAR** (Visual AutoRegressive), Parti, DALL-E 1.

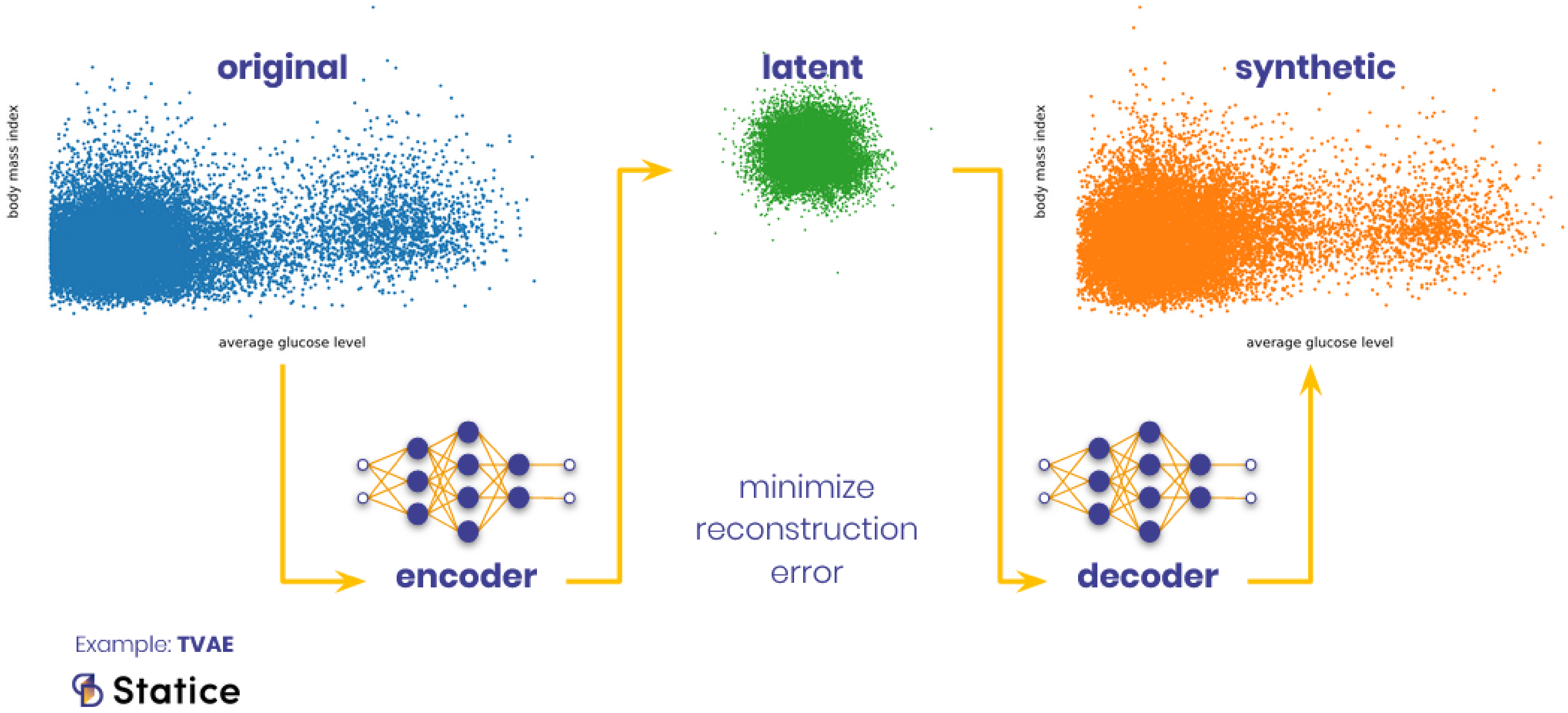
Variational Autoencoders





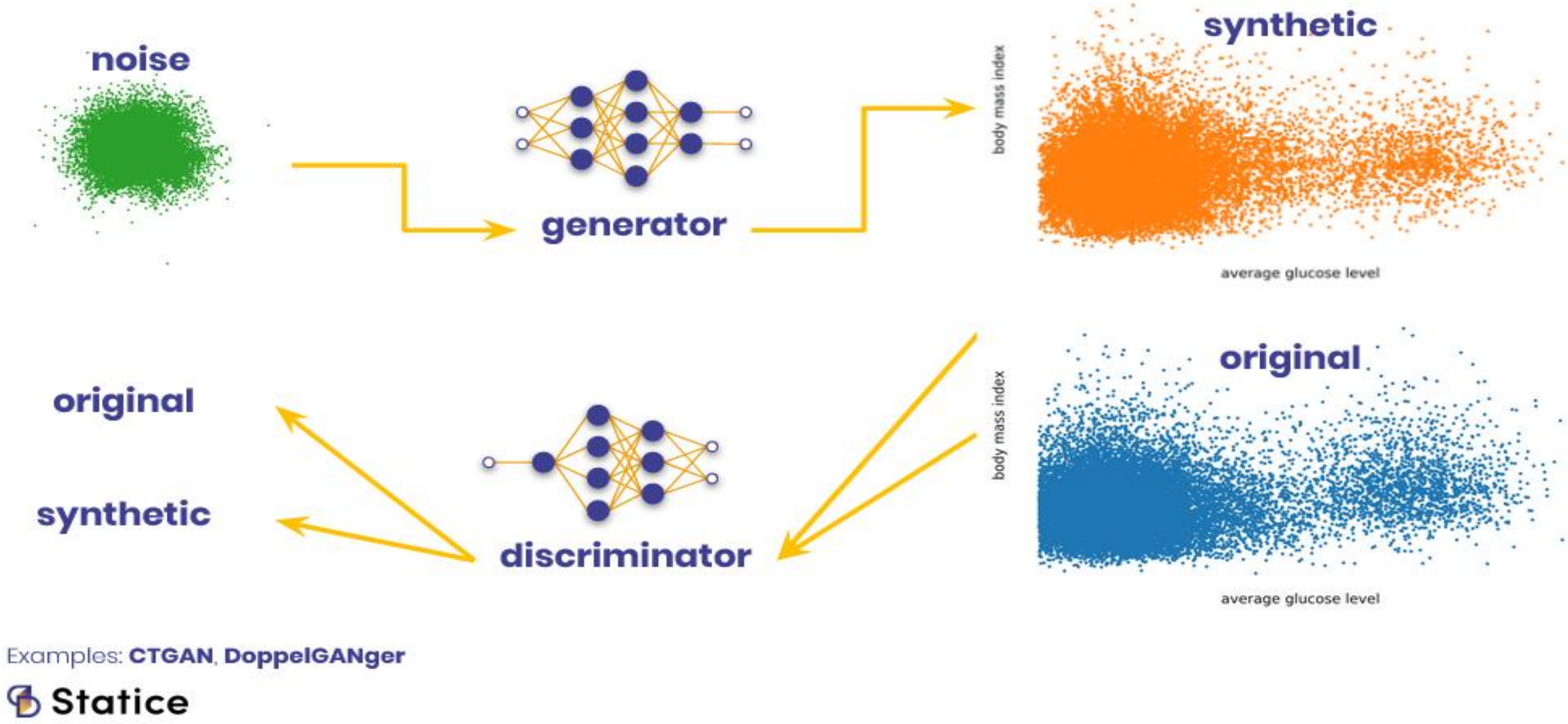
Source Credits: https://lilianweng.github.io/posts/2018-08-12-vae/

Variational Autoencoders



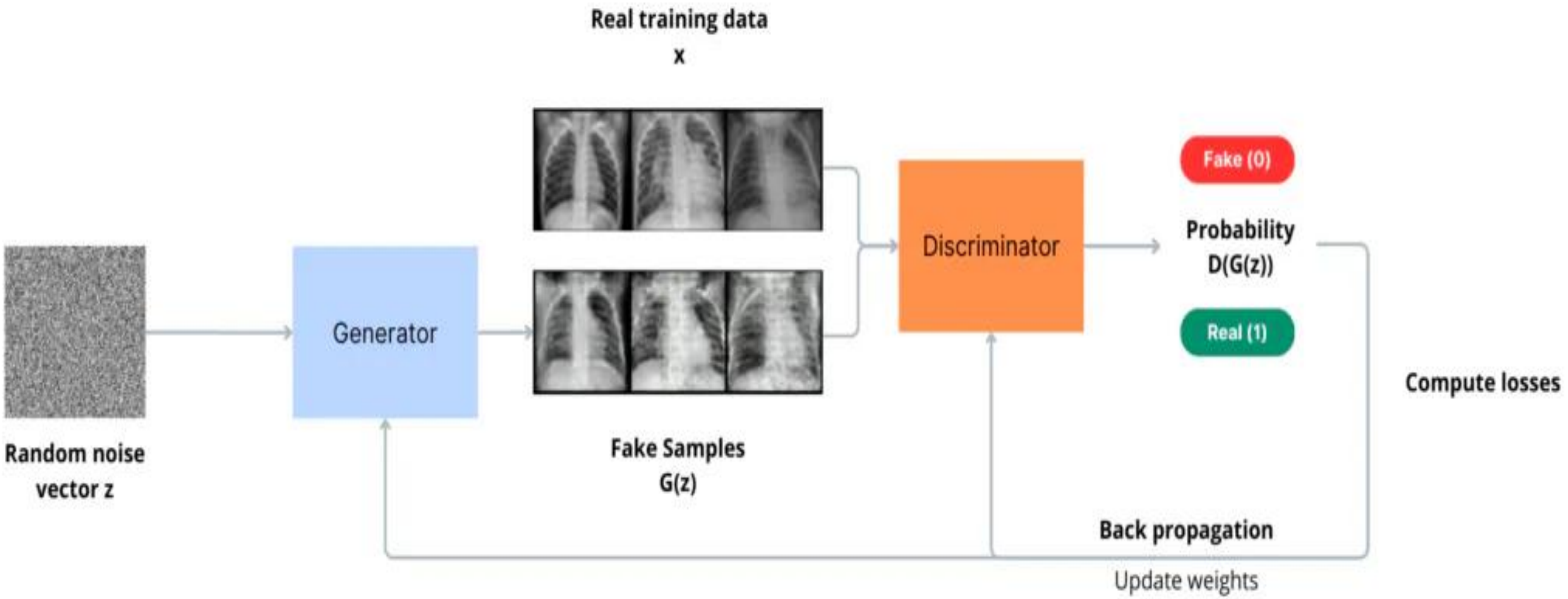
Source Credits: https://sungsoo.github.io/2021/07/10/synthetic-data.html

Generative Adversarial Networks (GANs)

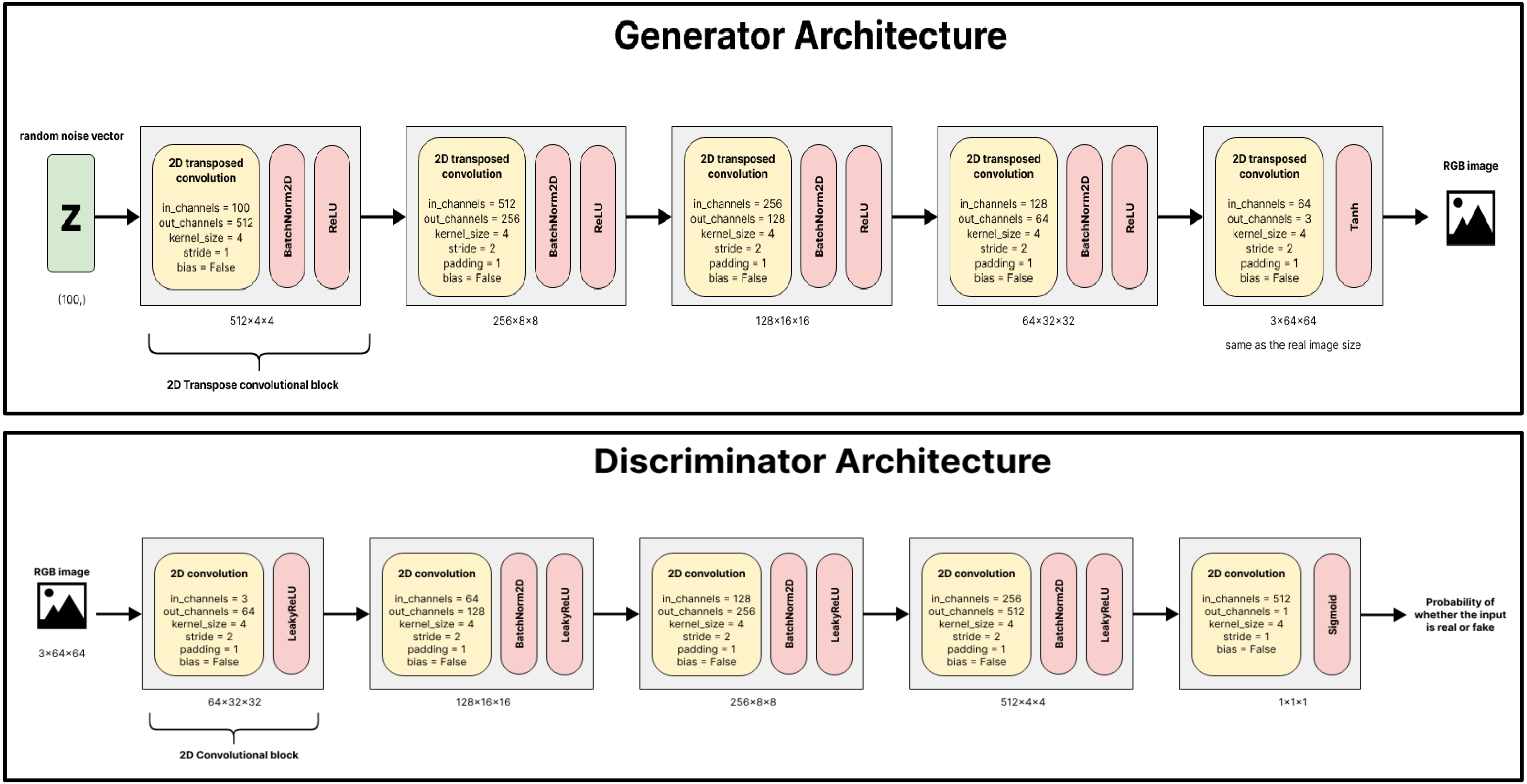


Source Credits: https://sungsoo.github.io/2021/07/10/synthetic-data.html

Generative Adversarial Networks (GANs)

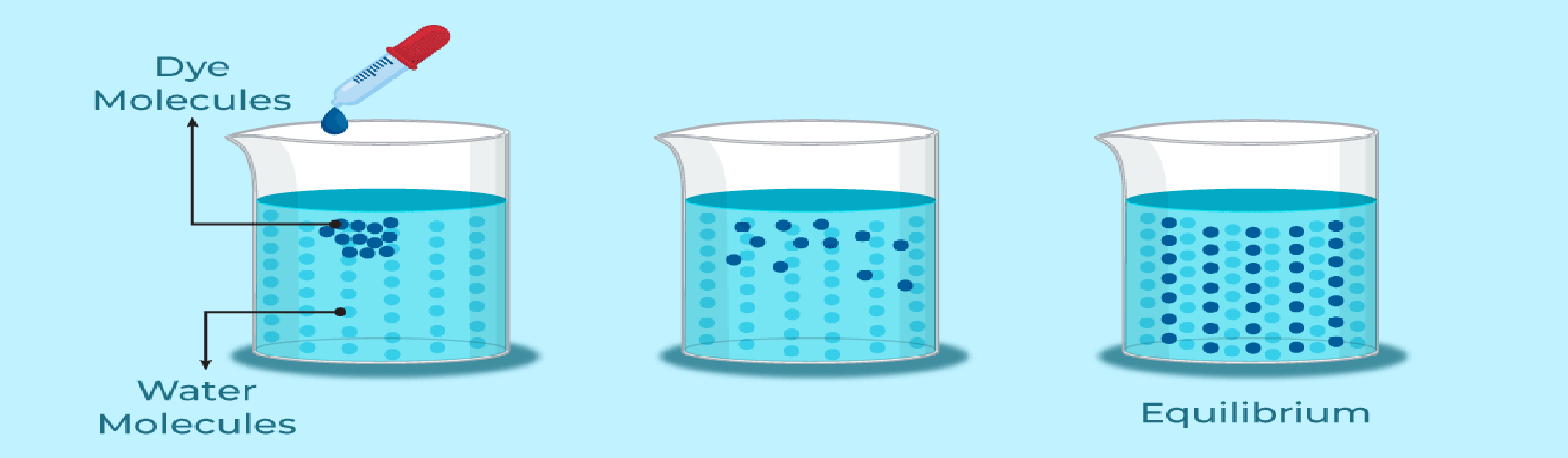


Source Credits: https://blog.ovhcloud.com/en/posts/understanding-image-generation-beginner-guide-generative-adversarial-networks-gan/



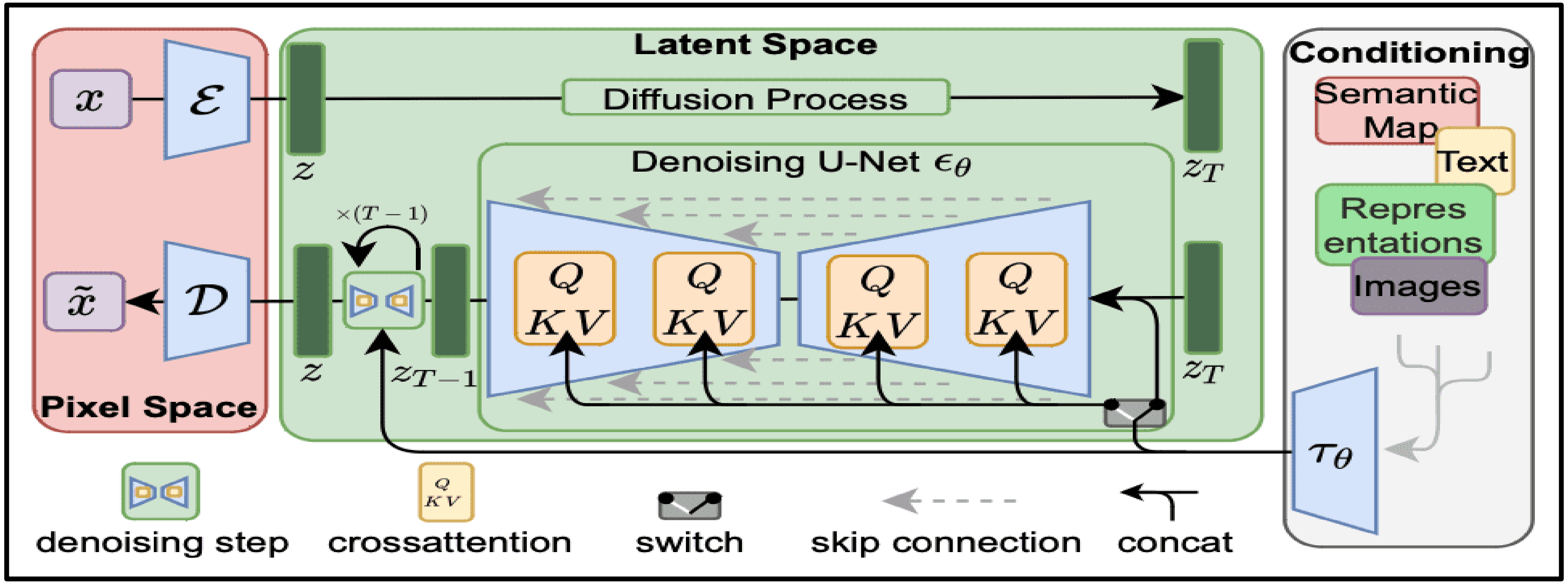
Source Credits: https://blog.ovhcloud.com/en/posts/understanding-image-generation-beginner-guide-generative-adversarial-networks-gan/

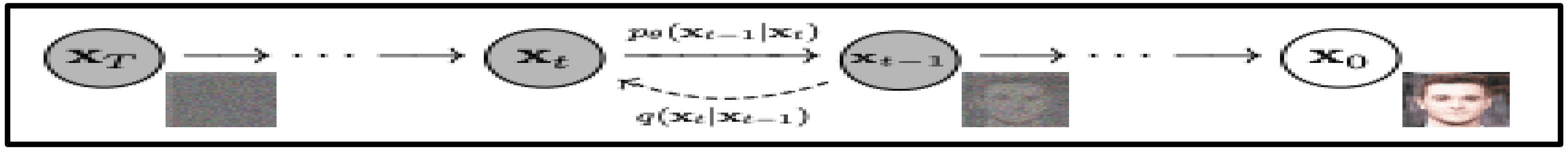
Diffusion



Source Credits: GeeksforGeeks

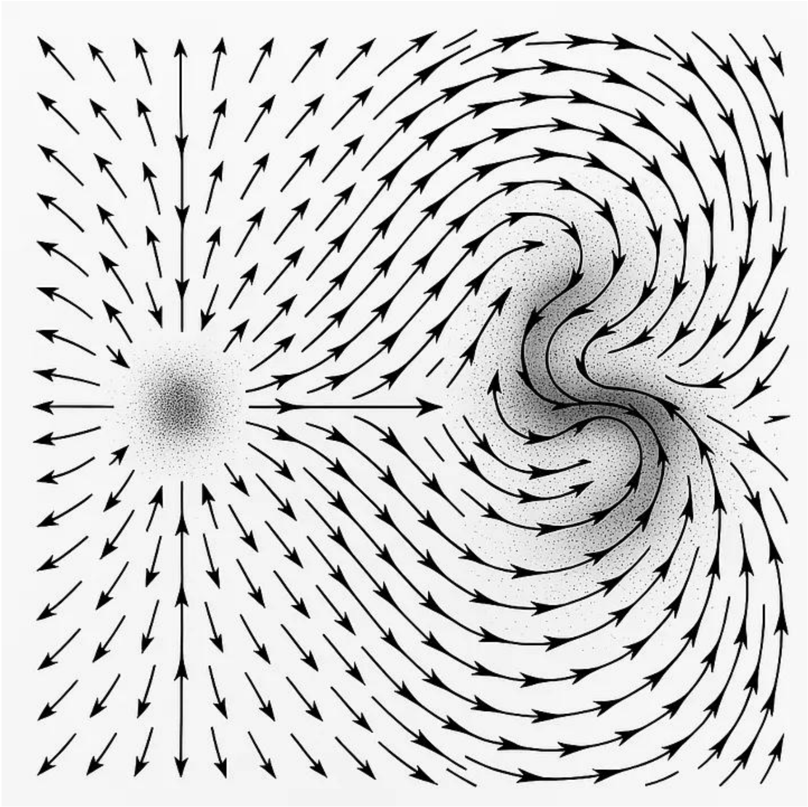
Diffusion Models





Source Credits: Rombach et al (2021)

Continuous Flow Matching



Unlike diffusion models that predict random noise, flow matching models train a neural network to predict a vector field v(x,t) that pushes data along a straight or optimal path

Source Credits: https://arxiv.org/pdf/2210.02747

Autoregressive Visual Models

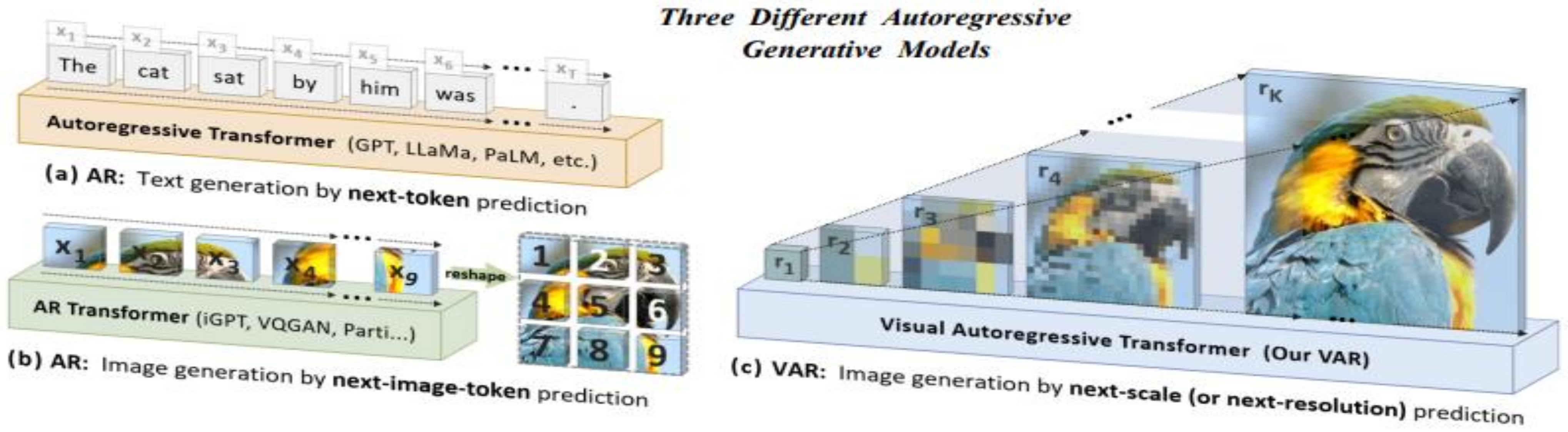
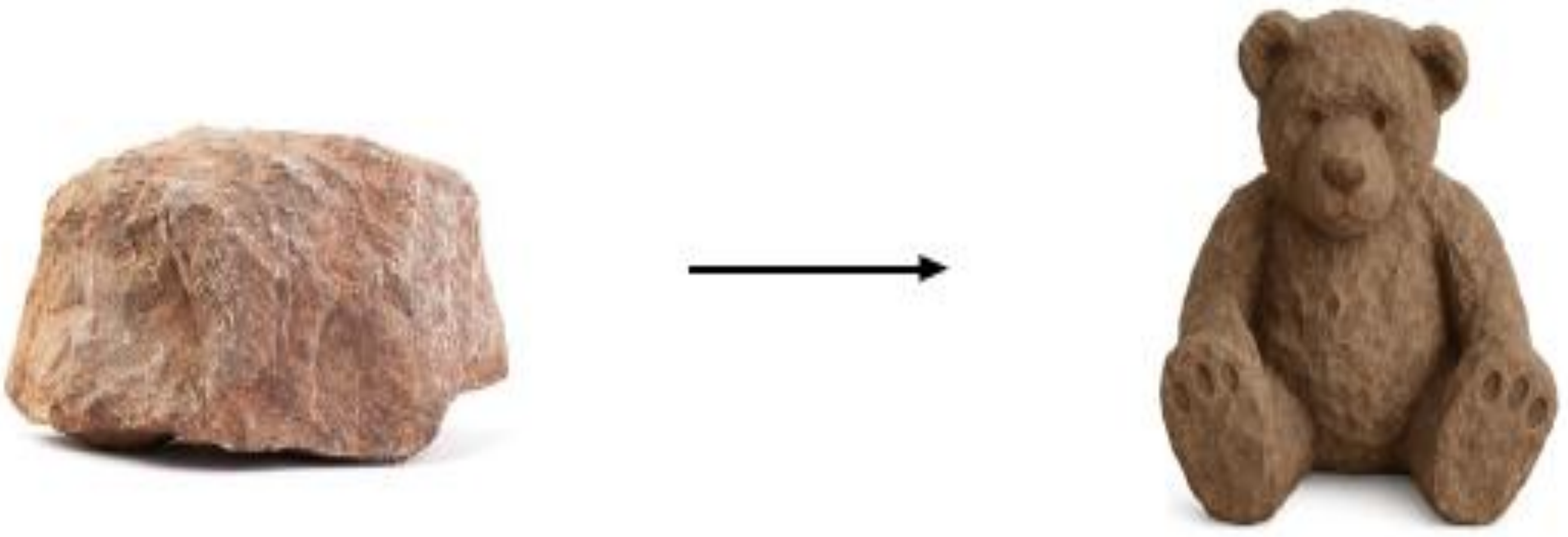


Figure 2: **Standard autoregressive modeling (AR)** ***vs.*** **our proposed visual autoregressive modeling (VAR).** (a) AR applied to language: sequential text token generation from left to right, word by word; (b) AR applied to images: sequential visual token generation in a raster-scan order, from left to right, top to bottom; (c) VAR for images: multi-scale token maps are autoregressively generated from coarse to fine scales (lower to higher resolutions), with parallel token generation within each scale. VAR requires a multi-scale VQVAE to work.

https://arxiv.org/pdf/2404.02905

Diffusion Based Models

Think about carving out a sculpture from a stone



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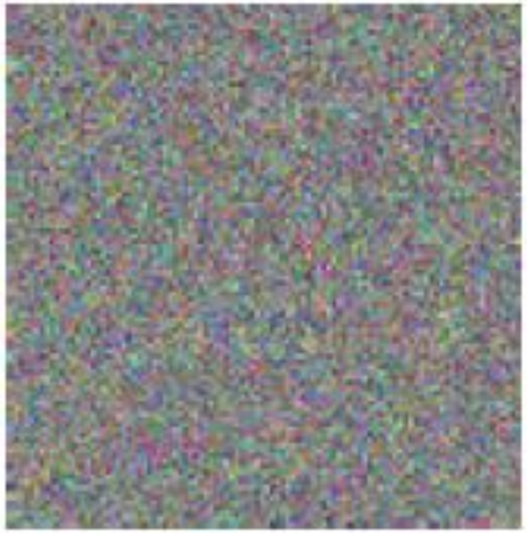
Diffusion : generate images from random noise



Why start from noise?

1. Noise is easy to sample
2. Introduces randomness
3. Gaussian distribution has nice properties

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Diffusion Based Models for Image Generation

- Forward Diffusion: adding random Gaussian noise to an image step by step until the image turns into pure, random noise.

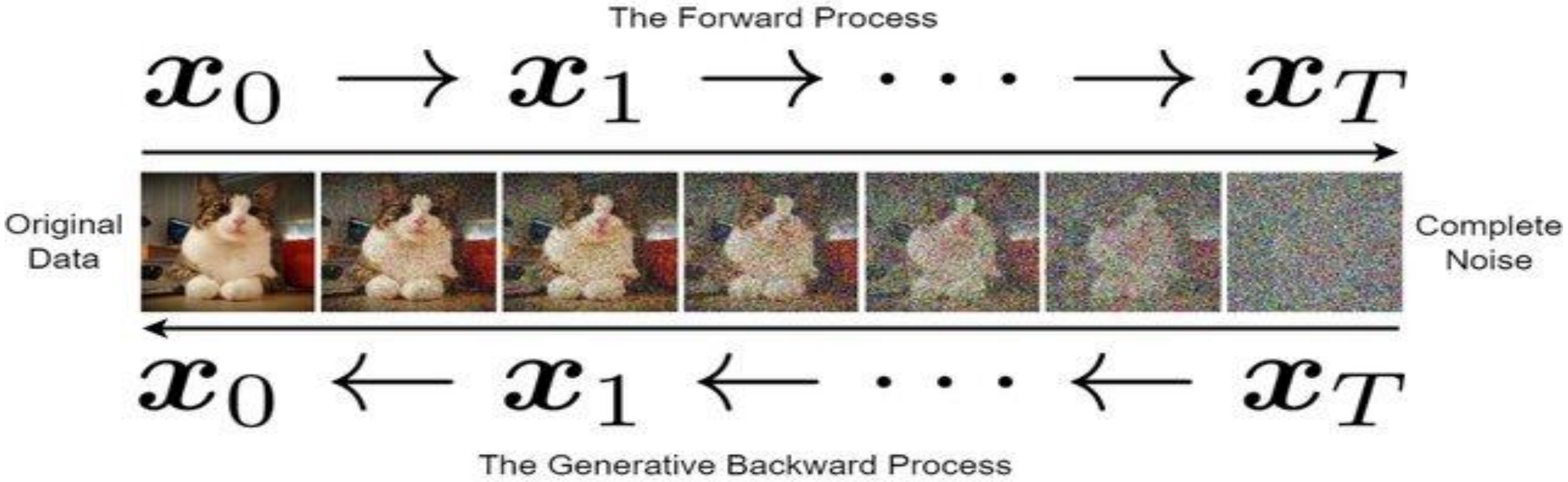
- Reverse Diffusion: Reverse the process of adding noise. Use trained neural networks to estimate and remove a tiny bit of noise at each step, moving backward from pure noise to a clear image.

- The model learns the "score function," which is the gradient of the data probability density, acting like a force field that pushes messy noise back into the shape of a recognizable image.

Diffusion for Image Generation

- **Diffusion for image generation** is a class of generative models that synthesize images by learning to reverse a two-phase Markov process: **adding noise to data and then removing it**.
  - **Forward Process (Noising):** A clean image x_0 is incrementally corrupted by adding Gaussian noise over T timesteps until it becomes pure thermal/random noise x_T.
  - **Reverse Process (Denoising):** A neural network (typically a U-Net or Transformer) is trained to predict and subtract the noise added at each step, generating a clean image starting from pure random noise.

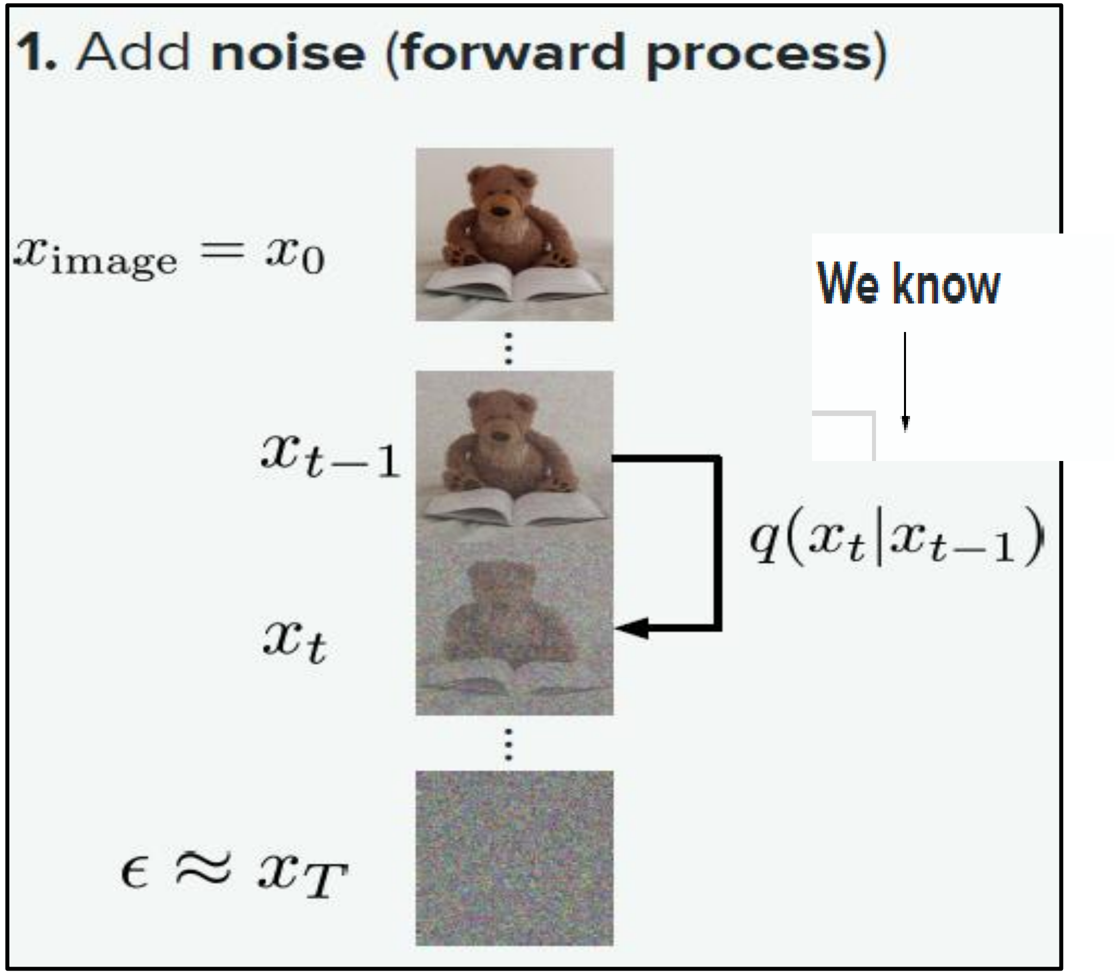
The forward and backward processes of the diffusion model.



In **conditioned diffusion** (such as Text-to-Image), text embeddings from a VLM or text encoder (e.g., CLIP, T5) are injected into the denoising network via cross-attention layers to guide the reverse process toward the visual concepts specified in the prompt.

Source Credits: https://www.researchgate.net/publication/382128283_Diffusion_Models_Tutorial_and_Survey/figures?lo=1

Intuition behind diffusion



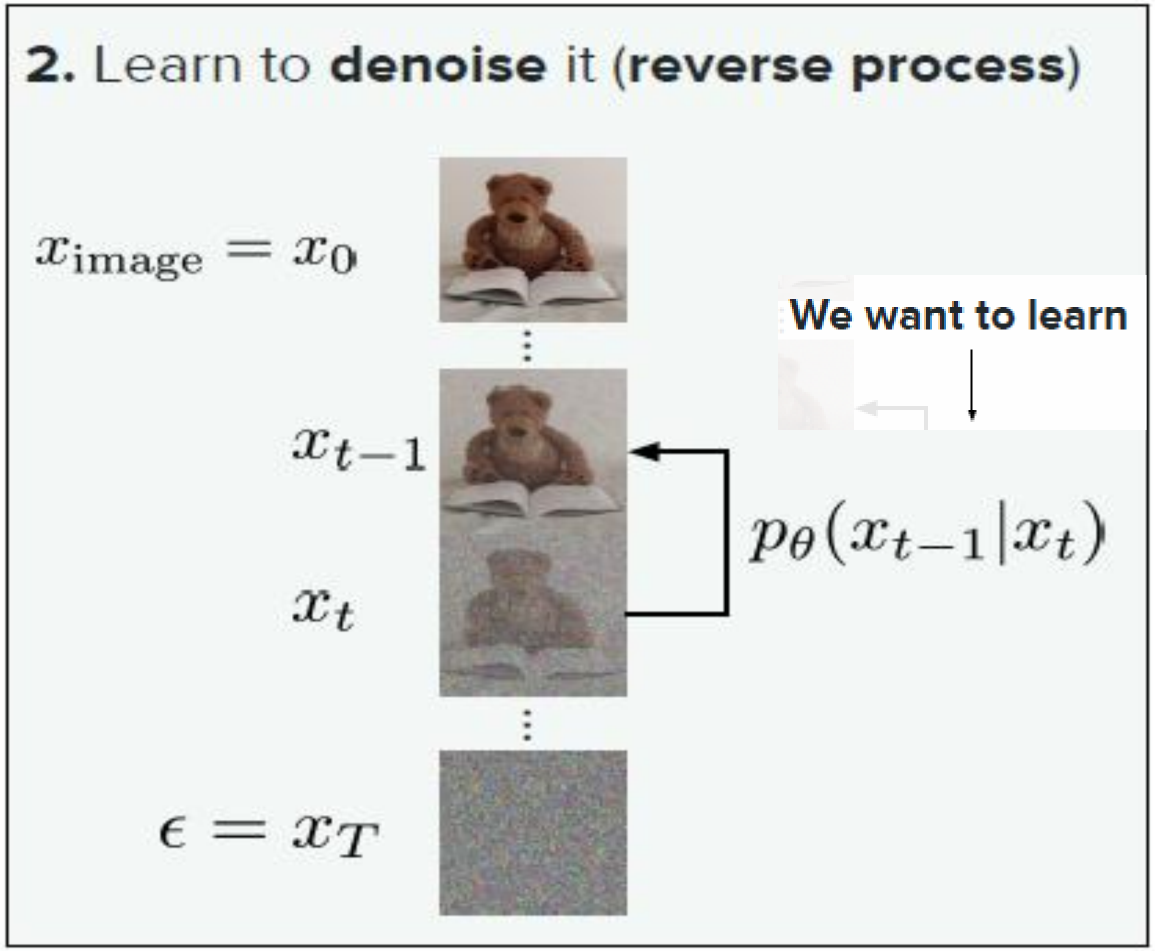


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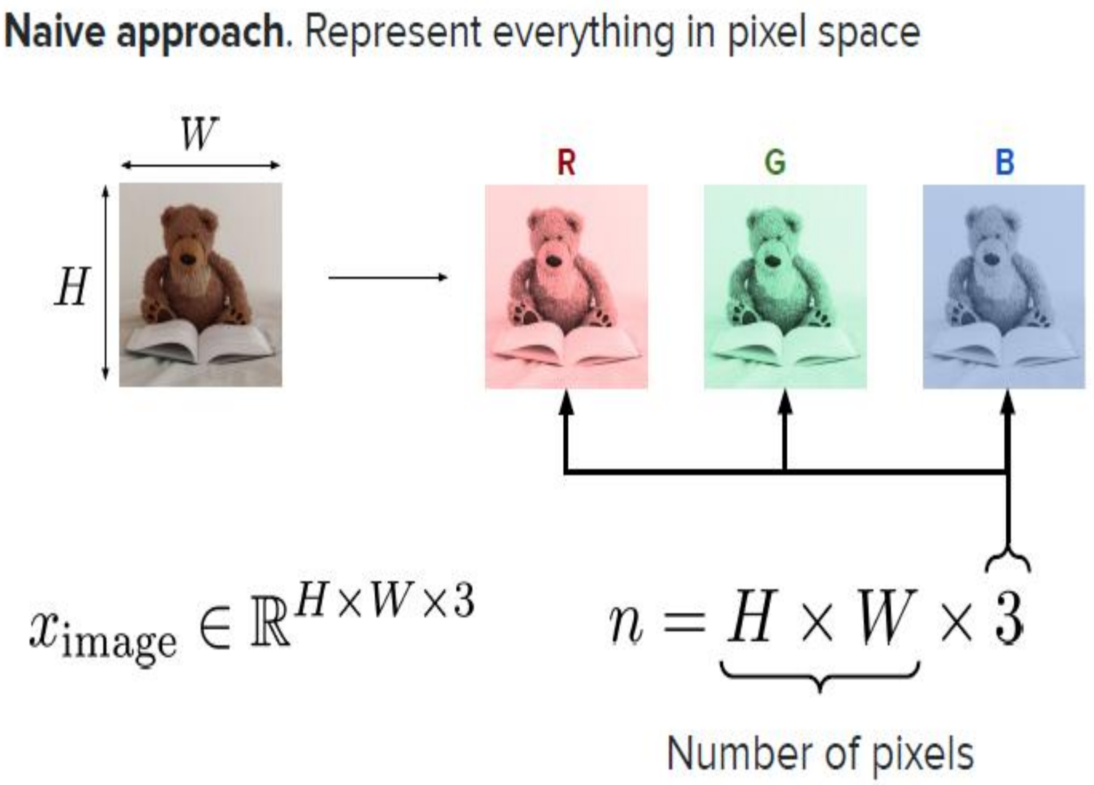
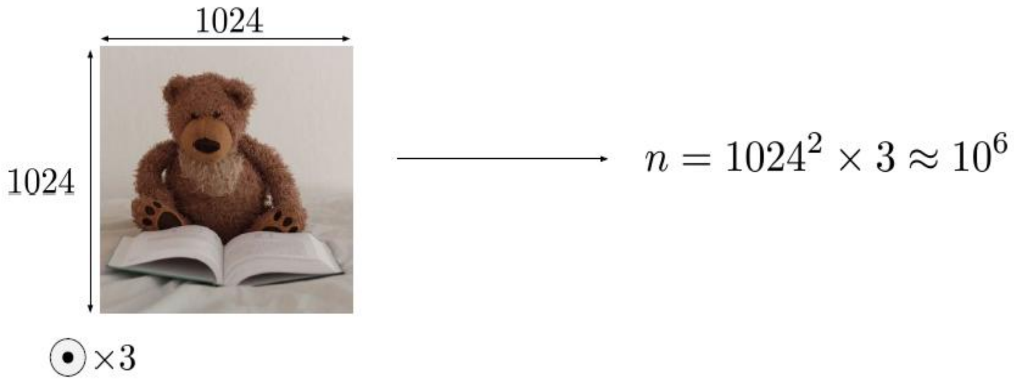
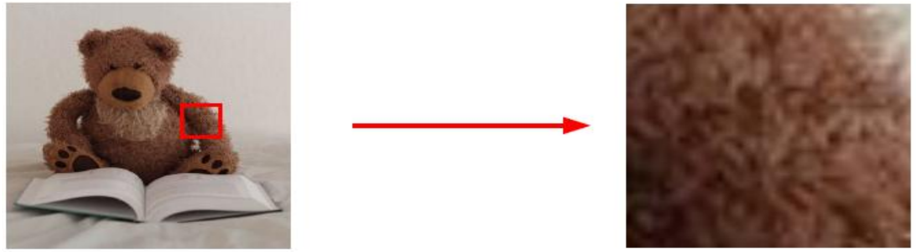
Image Representation in Pixel space







Limitations of Pixel Space Representation:
- Very high dimensional
- Redundant Information: Correlated Pixels
- Representation **not meaningful** (if we move in space, image becomes gibberish)

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Image Representation

$$\longleftrightarrow \quad x_{\text{image}} = \begin{pmatrix} x^{(1)} \\ x^{(2)} \\ \vdots \\ x^{(n)} \end{pmatrix}$$

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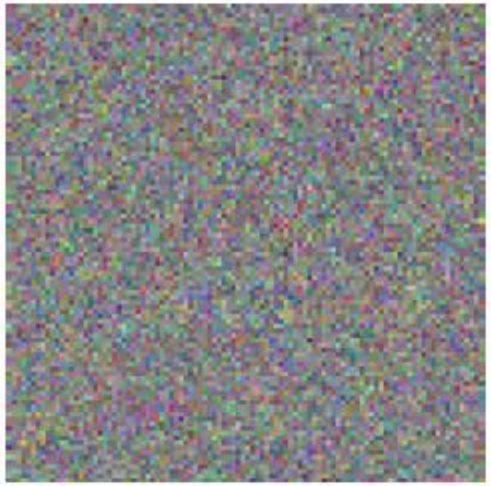
Noise representation



$$\longleftrightarrow \quad \epsilon = \begin{pmatrix} \epsilon^{(1)} \\ \epsilon^{(2)} \\ \vdots \\ \epsilon^{(n)} \end{pmatrix} \sim \mathcal{N}(0, I)$$

In other words, $\epsilon^{(i)} \sim \mathcal{N}(0,1)$ independent and identically distributed

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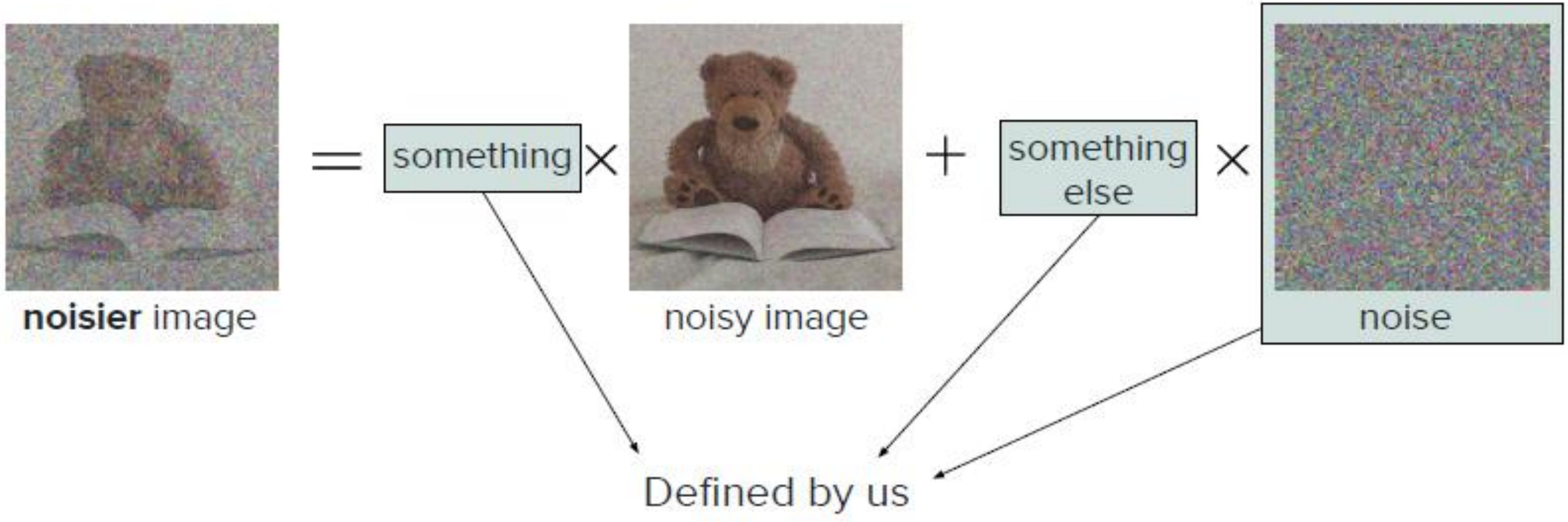
Process behind Diffusion

q is **chosen** by us and its **distribution** is **known** (reminder: it is stochastic!)



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Kullback-Leibler (KL) Divergence loss

- In diffusion models, the **Kullback-Leibler (KL) Divergence loss** measures how closely the model's predicted reverse step $p_\theta(x_{t-1}|x_t)$ matches the true posterior distribution $q(x_{t-1}|x_t, x_0)$.

https://haroldbenoit.com/notes/ml/diffusion-models/frameworks---theory/variational-diffusion

Paper

Denoising Diffusion Probabilistic Models

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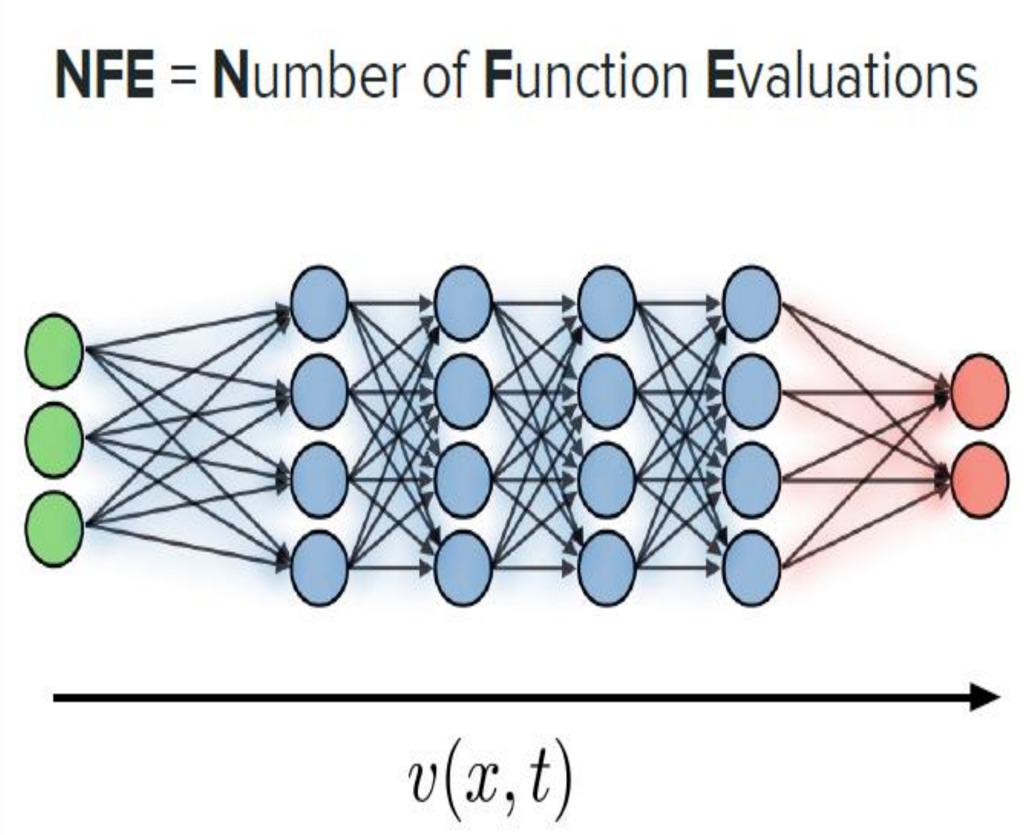
Abstract

We present high quality image synthesis results using diffusion probabilistic models, a class of latent variable models inspired by considerations from nonequilibrium thermodynamics. Our best results are obtained by training on a weighted variational bound designed according to a novel connection between diffusion probabilistic models and denoising score matching with Langevin dynamics, and our models naturally admit a progressive lossy decompression scheme that can be interpreted as a generalization of autoregressive decoding. On the unconditional CIFAR10 dataset, we obtain an Inception score of 9.46 and a state-of-the-art FID score of 3.17. On 256x256 LSUN, we obtain sample quality similar to ProgressiveGAN. Our implementation is available at `https://github.com/hojonathanho/diffusion`.

Complexity metric: NFE



- **Number of Function Evaluations (NFE)** is an inference computational complexity metric that measures **the total number of forward passes through a neural network (e.g., U-Net or Transformer) required to generate a single sample.**

- In diffusion and flow-matching models, generating an image involves integrating a continuous **Ordinary Differential Equation (ODE) or Stochastic Differential Equation (SDE) backwards** in time across T discretization steps.

- NFE measures the actual model evaluations performed during this integration.

Flow Matching Based Generative Models

Flow Matching generation

- State-of-the-art models like Flux and Stable Diffusion 3 use flow matching.

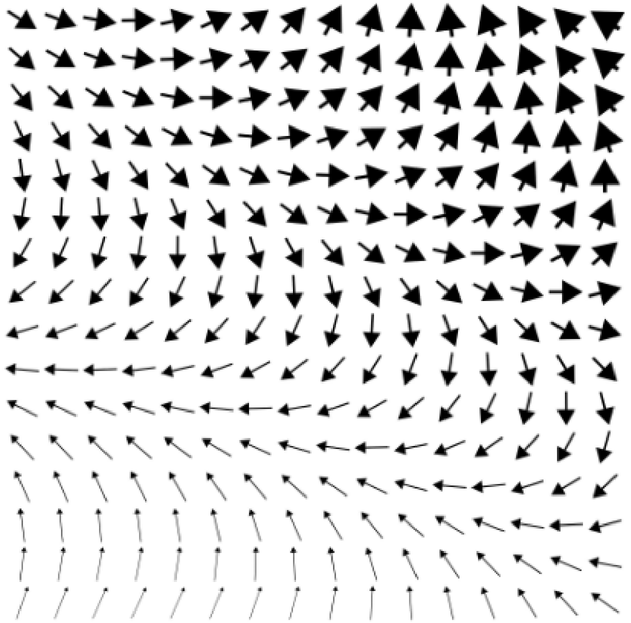
- They define a straight, continuous path (a "flow") to smoothly morph a grid of random Gaussian noise into a crisp, clean image.

- The neural network is trained to learn the "velocity" required to push those pixel values along that path efficiently

A Vector Field

- A **vector field** is a mathematical construct that assigns a vector to every point in a given spatial or temporal domain.

- Mathematically, a vector field in n-dimensional space is defined as a function $v: \mathbb{R}^n \to \mathbb{R}^n$(or time-dependent $v: \mathbb{R}^n \times [0, T] \to \mathbb{R}^n$).



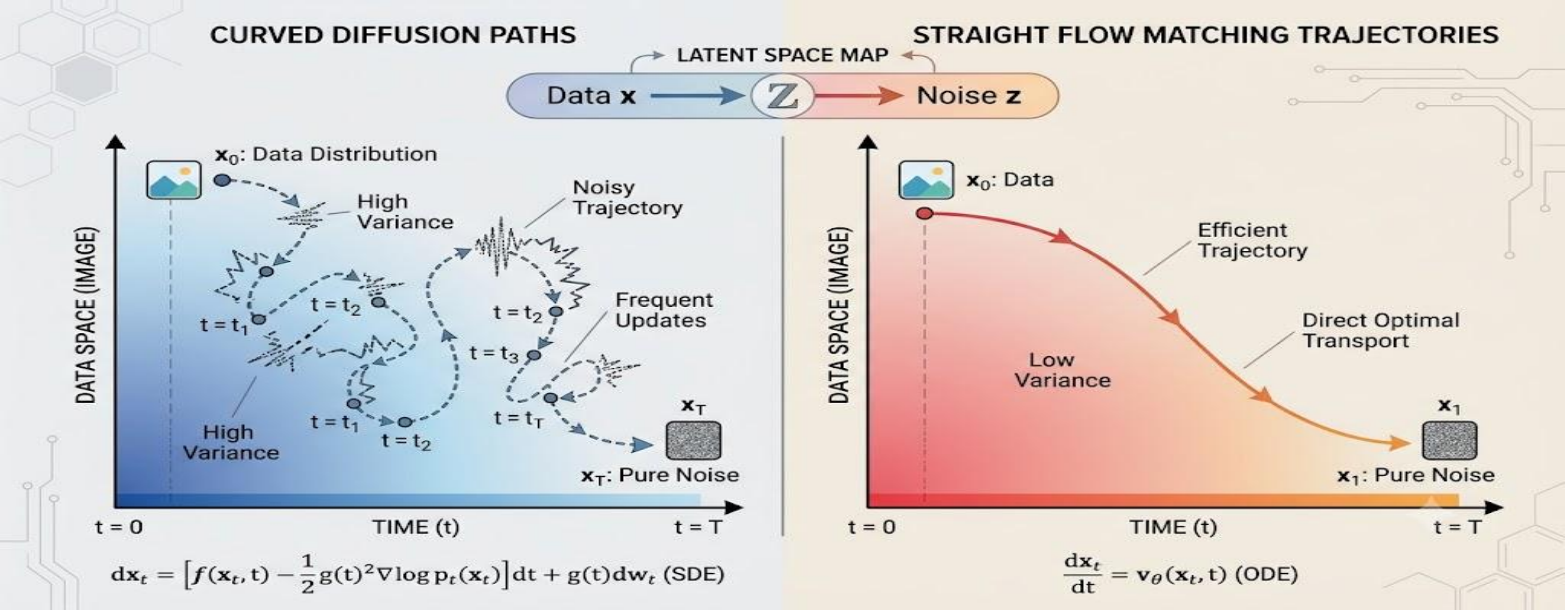
Vector fields in Generative Modelling

In continuous generative models (Continuous Normalizing Flows, Score-Based Models, and Flow Matching), the image space is treated as a high-dimensional continuous domain.

- **Velocity Field (v_θ):** A neural network learns a time-dependent vector field $v_\theta(x_t, t)$that dictates how points move smoothly through continuous latent space.
- **ODE Trajectories:** Generating an image corresponds to starting at a random noise point $x_0 \sim \mathcal{N}(0, \mathbf{I})$and pushing it along the arrows of the learned vector field over time $t \in [0,1]$by solving the Ordinary Differential Equation:

$$\frac{dx_t}{dt} = v_\theta(x_t, t)$$

- **Diffusion:** Generates curved, noisy vector fields due to Brownian motion (SDEs), requiring many integration steps (high NFE)
- **Flow Matching:** Explicitly trains the network to output **straight-line vector fields**, enabling fast trajectory solving in very few steps (NFE $= 4 - 20$).



Curved Diffusion Paths in Diffusion Paths

Straight Flow Matching Trajectories in Flow Based Models

Flow-based generative models

- **Flow-based generative models** are a class of exact likelihood-based generative architectures that transform a simple prior distribution (like a standard Gaussian $\mathcal{N}(0, \mathbf{I})$) into a complex data distribution (like natural images) using an **invertible and differentiable mapping** f_θ.

Continuous Flow Matching (Rectified Flow / Optimal Transport Flow)

- **Straight-Path Vector Fields:** Instead of curved trajectories like standard diffusion, Continuous Flow Matching learns a straight vector field connecting noise $x_0 \sim \mathcal{N}(0, \mathbf{I})$directly to data $x_1 \sim p_{\text{data}}$.

- **High Efficiency:** Straight paths allow sampling with extremely low NFE (NFE $= 4 - 20$) without complex solvers.

- **State of the Art:** Models like **FLUX.1**, **Stable Diffusion 3**, and **SDXL Turbo** rely on Flow Matching formulations rather than traditional diffusion process trajectories.

Multimodal Guided Generation

Multimodal guided generation

- **Multimodal guided generation** is the process of steering (guiding) a generative model using guidance signals from multiple distinct modalities simultaneously, such as text descriptions, reference images, depth maps, audio tracks, or spatial masks.

- **Joint Cross-Attention:** Conditional features from different input modalities (e.g., text prompt embeddings from T5/CLIP + visual embeddings from a reference image encoder) are concatenated or fed through separate cross-attention layers in the underlying neural network backbone (U-Net or Transformer).

- **Classifier-Free Guidance (CFG) / Score Combination:** The score vectors or predicted noise outputs from individual modalities are combined linearly to guide the reverse sampling process

Semantic similarity

- Structural
- Global geometry
- "Low" frequency

Perceptual similarity

- Local
- Texture
- "High" frequency

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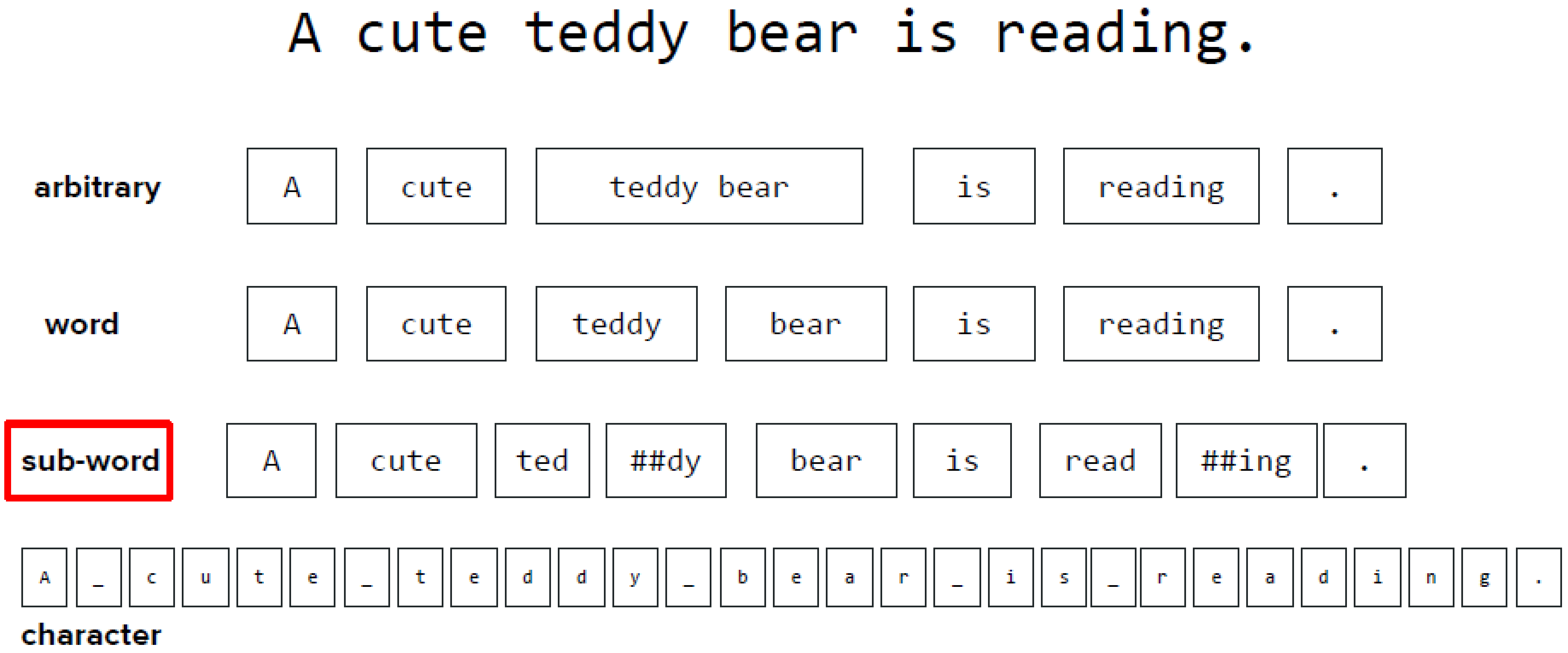
Multimodal Guided Generation (Image generation using Text Description): Tokenization



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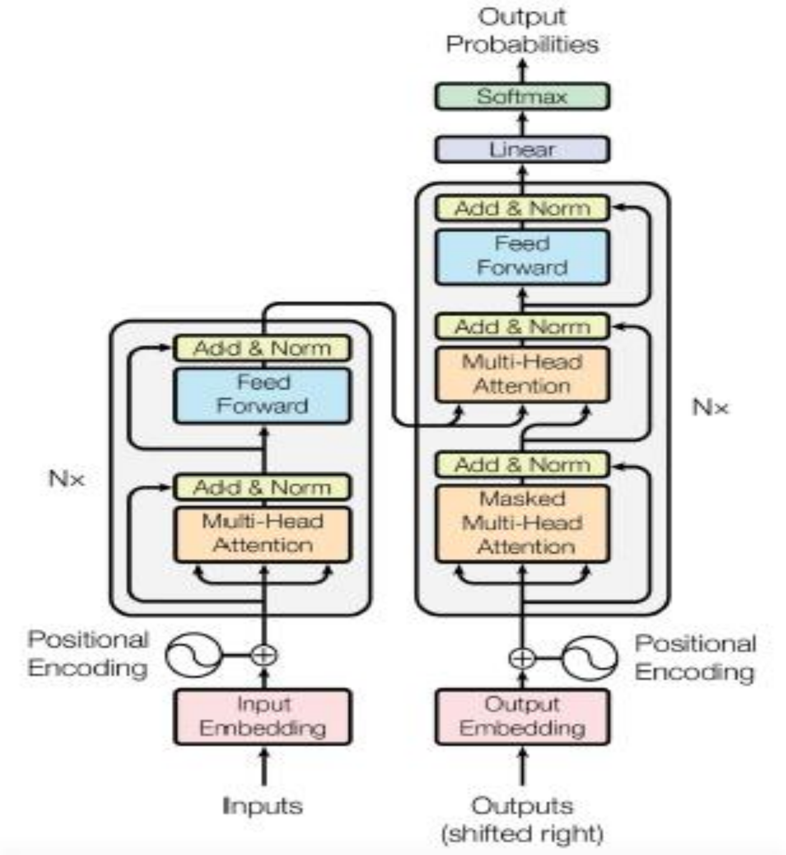
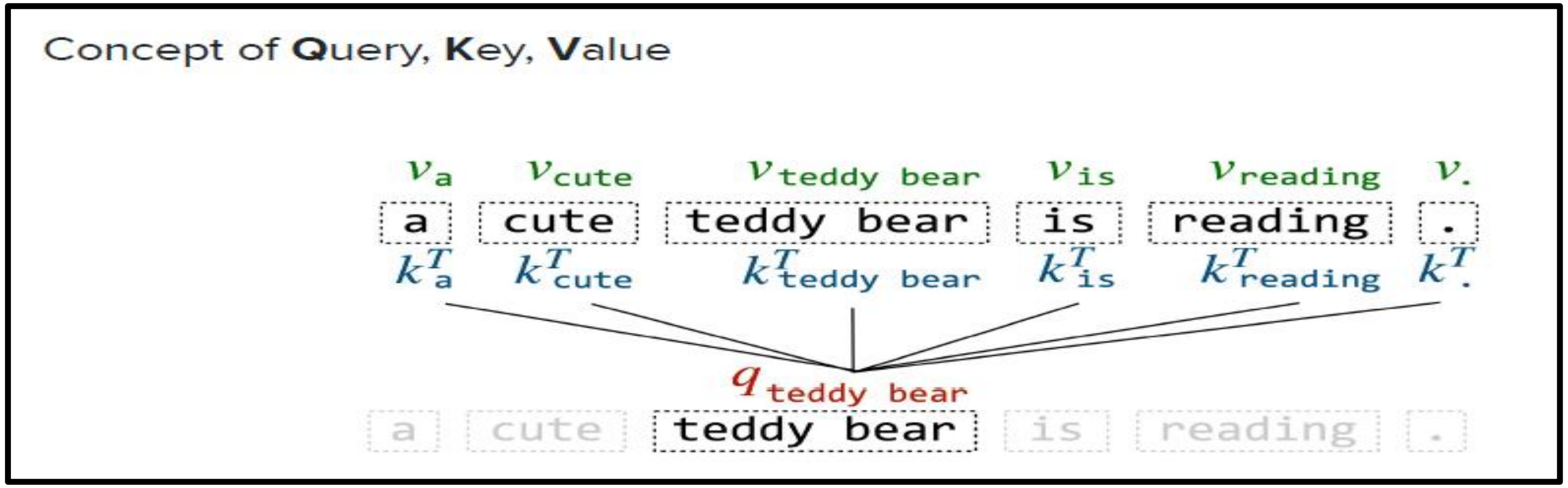
Transformer architecture

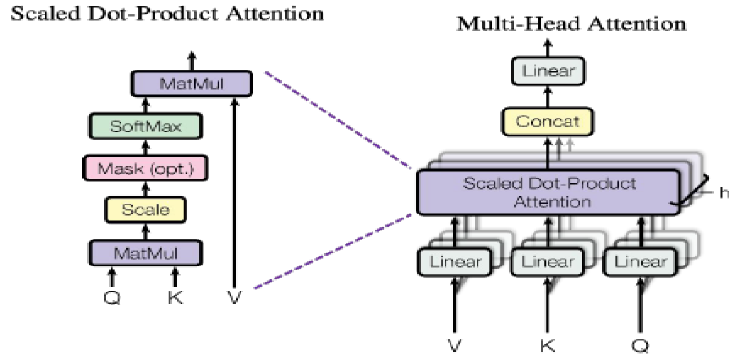




Attention Mechanism

Efficient computations with matrices:

$$\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



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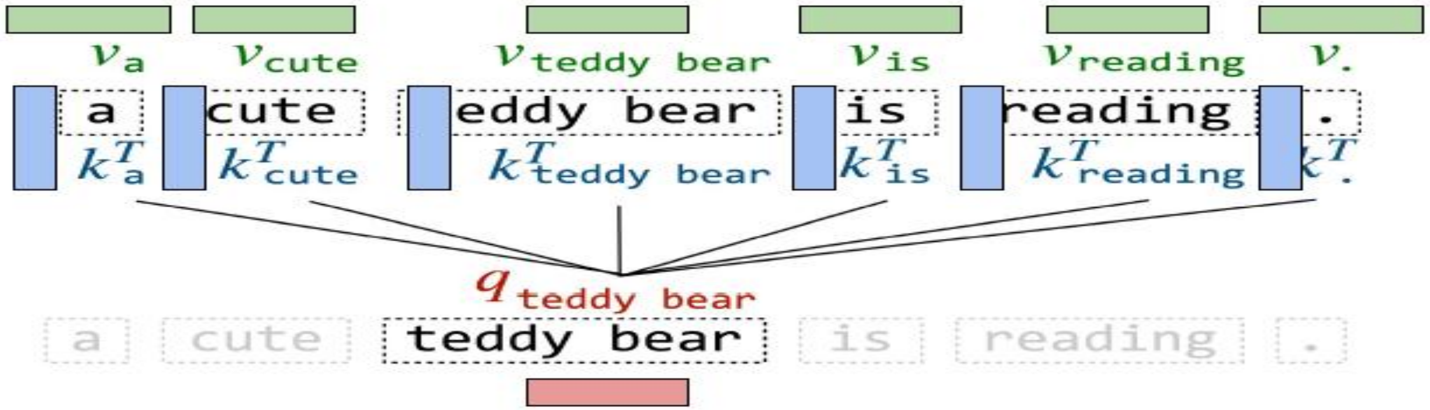
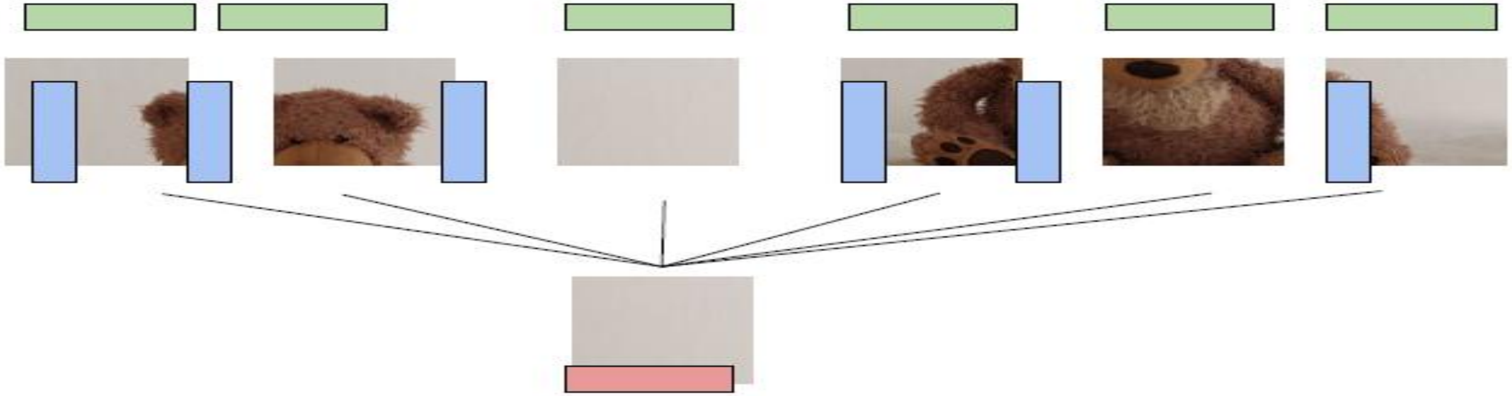
Adapting attention mechanism to images

It's just **numbers**...



...that could well represent **images**!



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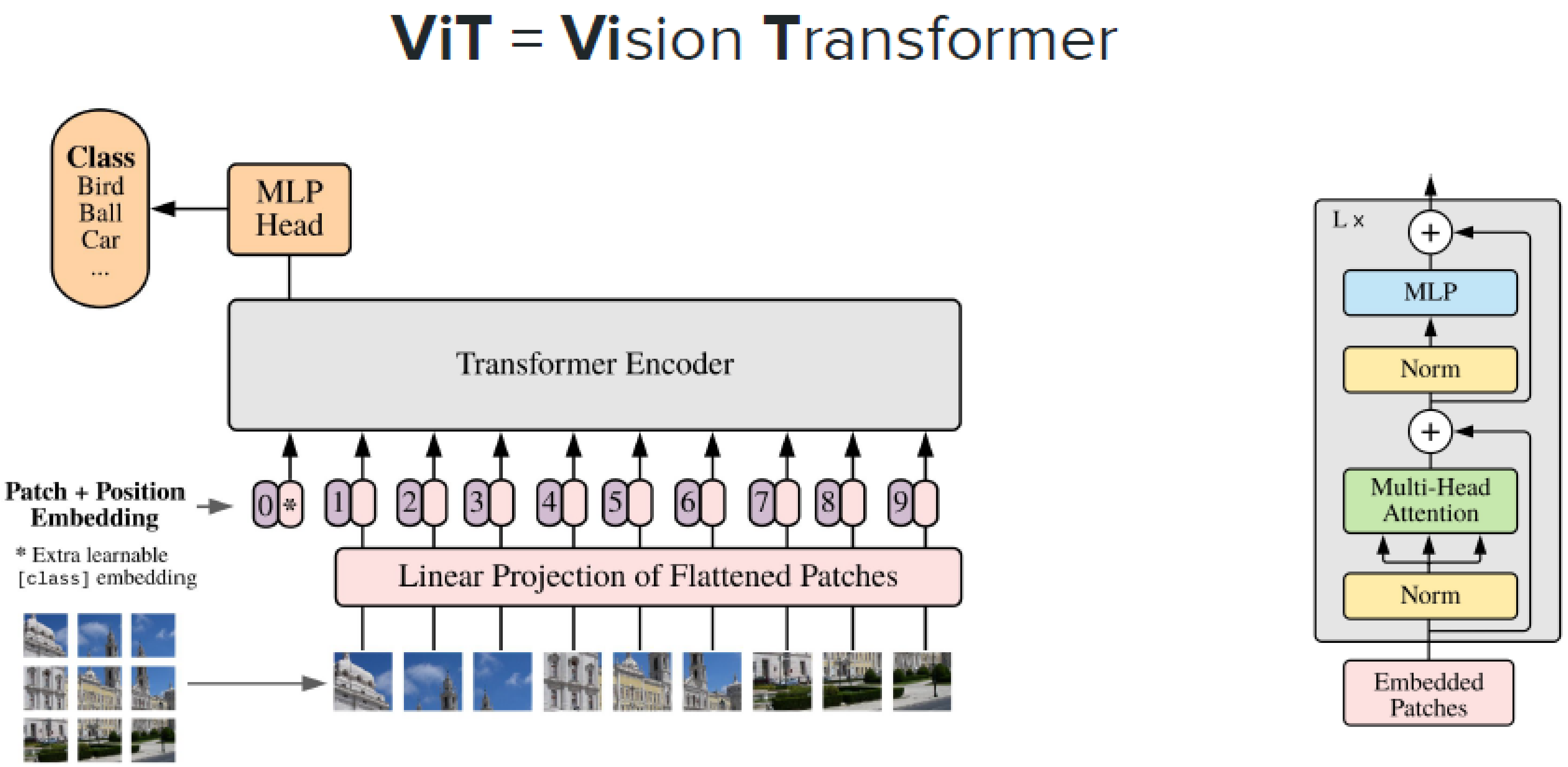
Vision Transformer



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Contrastive Learning: How to make image and text embeddings comparable?

- Contrastive learning: How to learn general image/text relationships?
- **Idea**. Contrastive learning:
  - Group similar items together
  - Push away dissimilar items

	teddy bear	water polo ball
	high similarity	low similarity
	low similarity	high similarity

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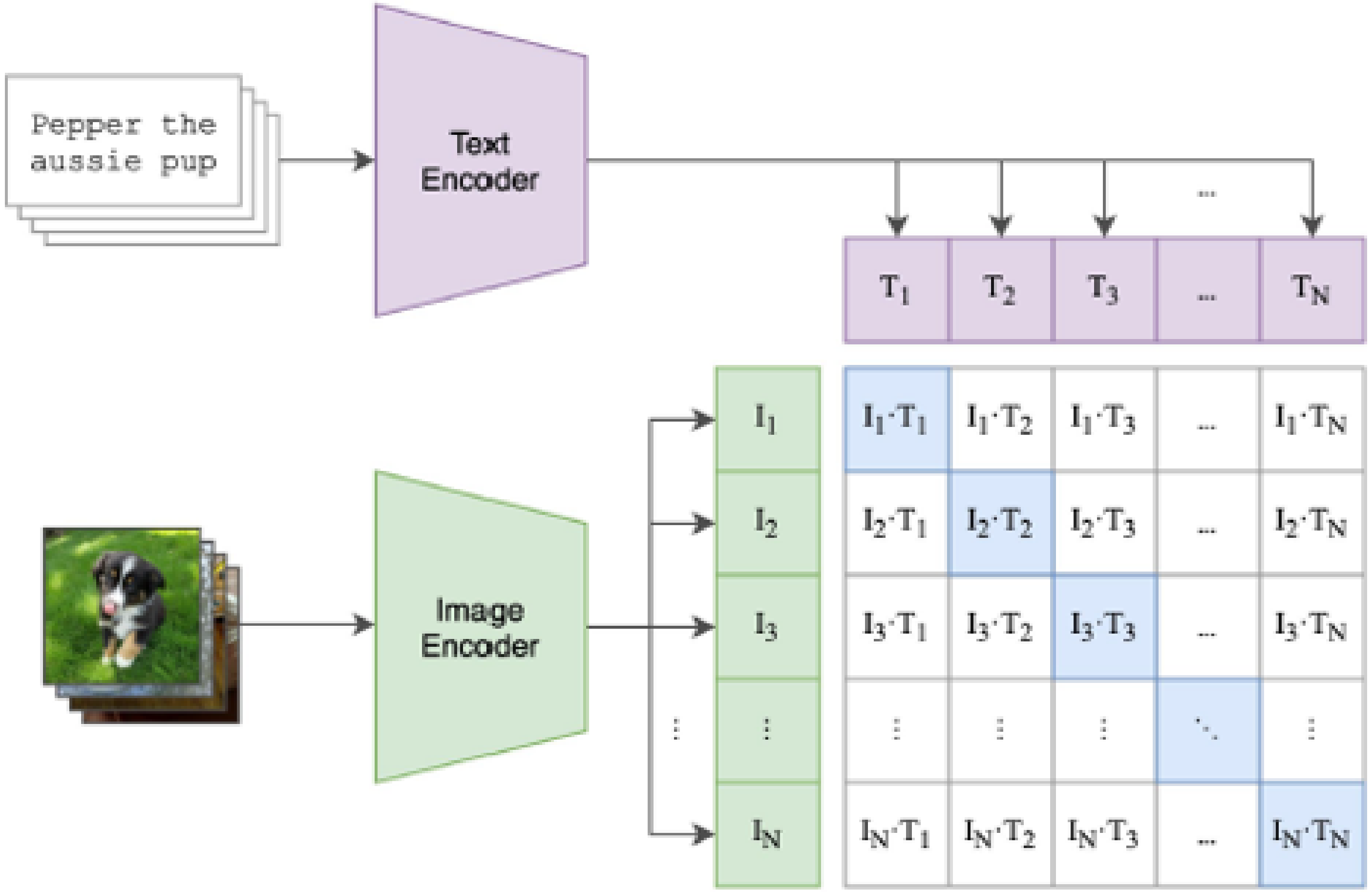
CLIP: Contrastive Language-Image Pretraining



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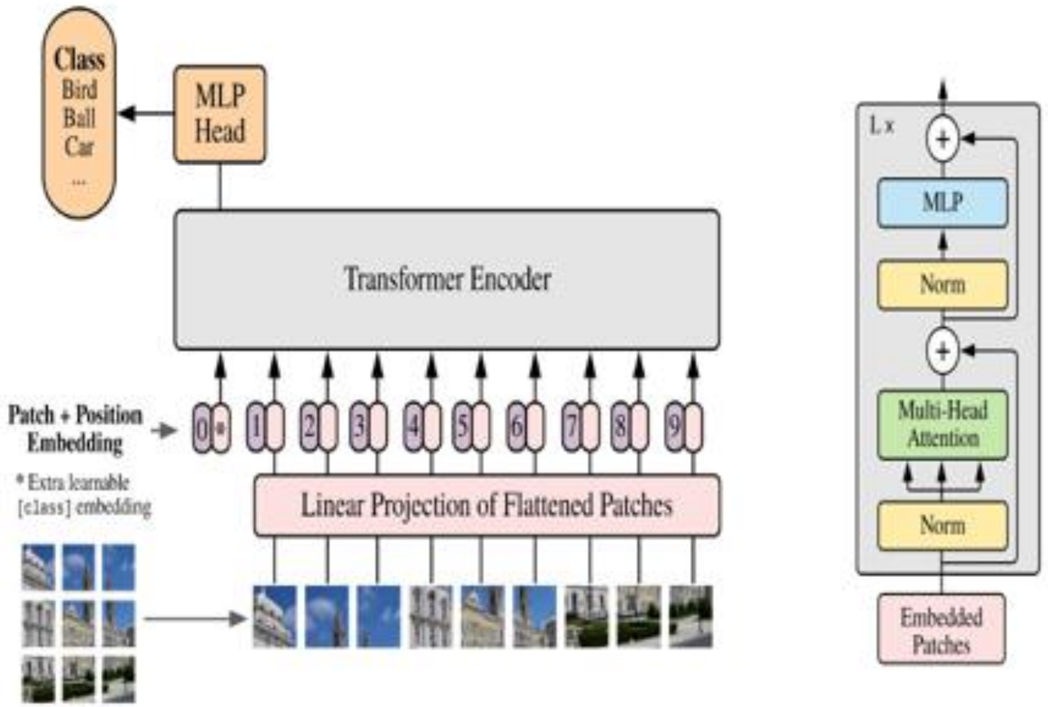
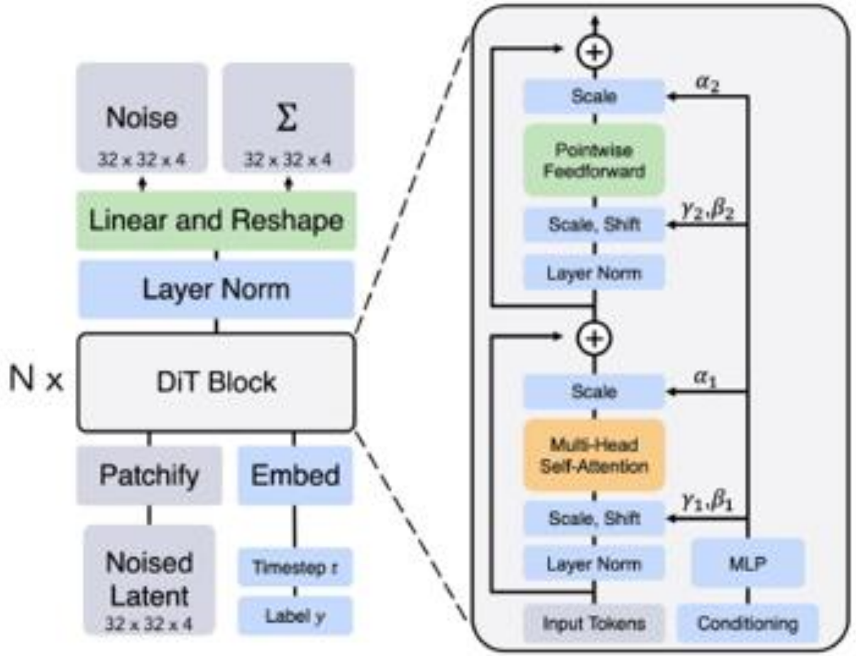
ViT = Vision Transformer



Transformer for vision understanding

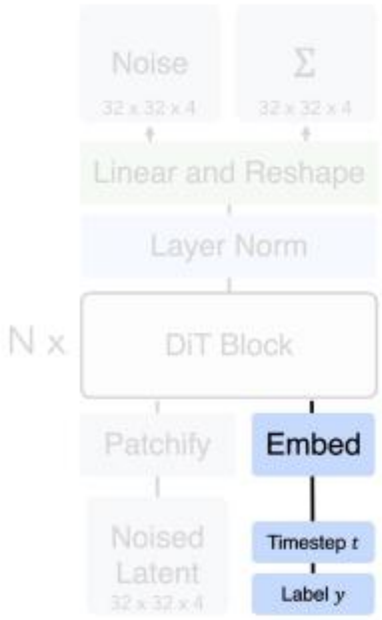
DiT = Diffusion Transformer



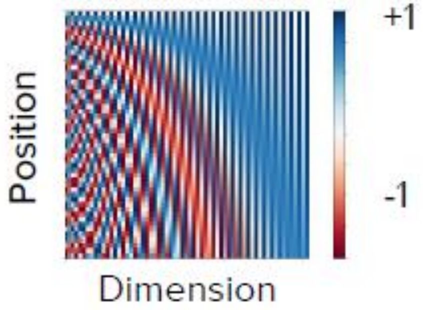
Transformer for vision generation

Source Credits: https://cme296.stanford.edu/

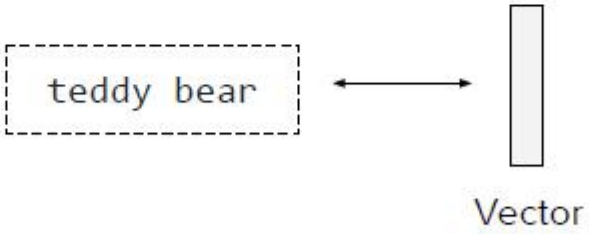
DiT: embedding of conditions



Timestep
embedding



Label
embedding



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Multi-Modal-Diffusion Transformer

MM-DiT = **M**ulti**M**odal**-Di**ffusion **T**ransformer

- Term coined in the **Stable Diffusion 3** paper published in 2024
- Relies on **joint attention** of different modalities

Single-stream
Everyone treated equally

Hybrid
Contains both "single-stream" and "double-stream" layer

Double-stream
Each modality is in its own stream

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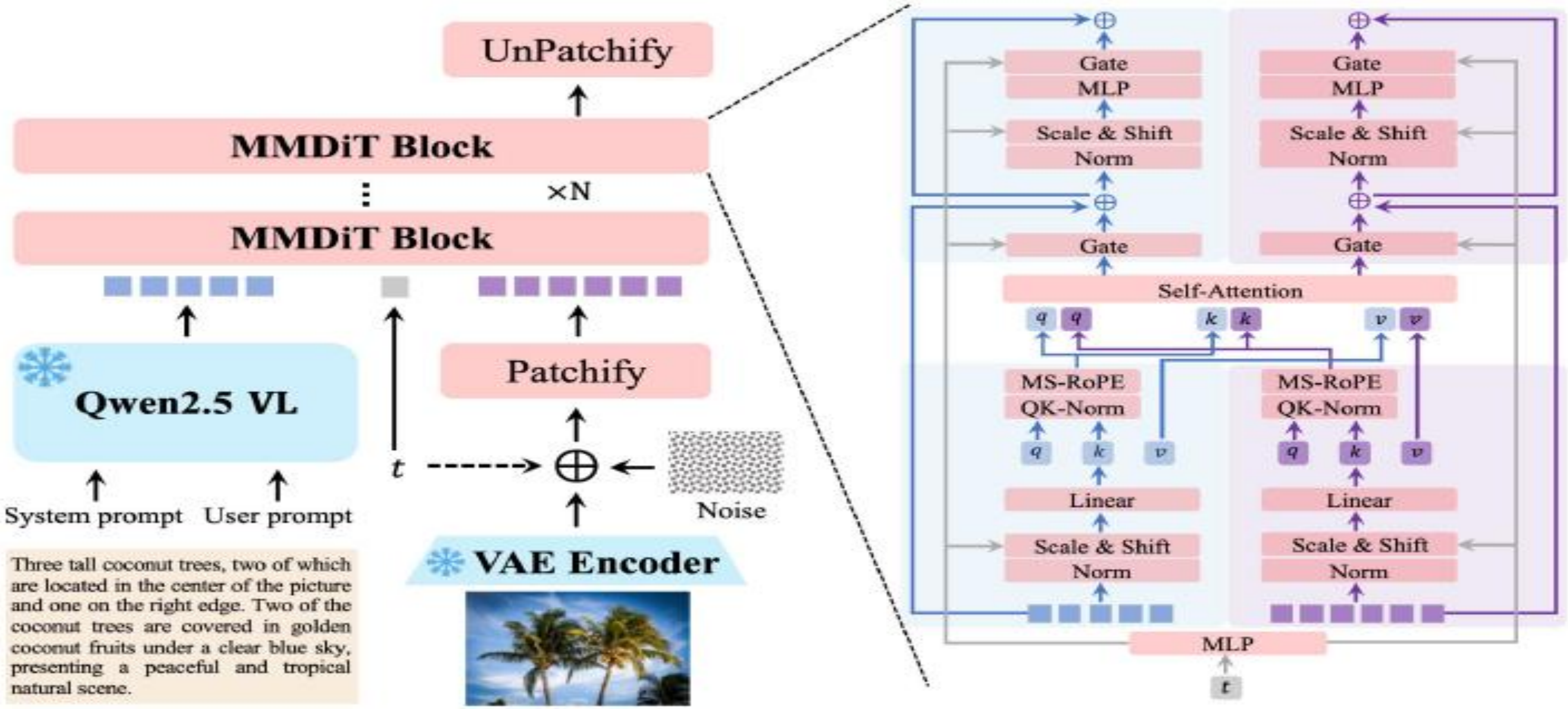
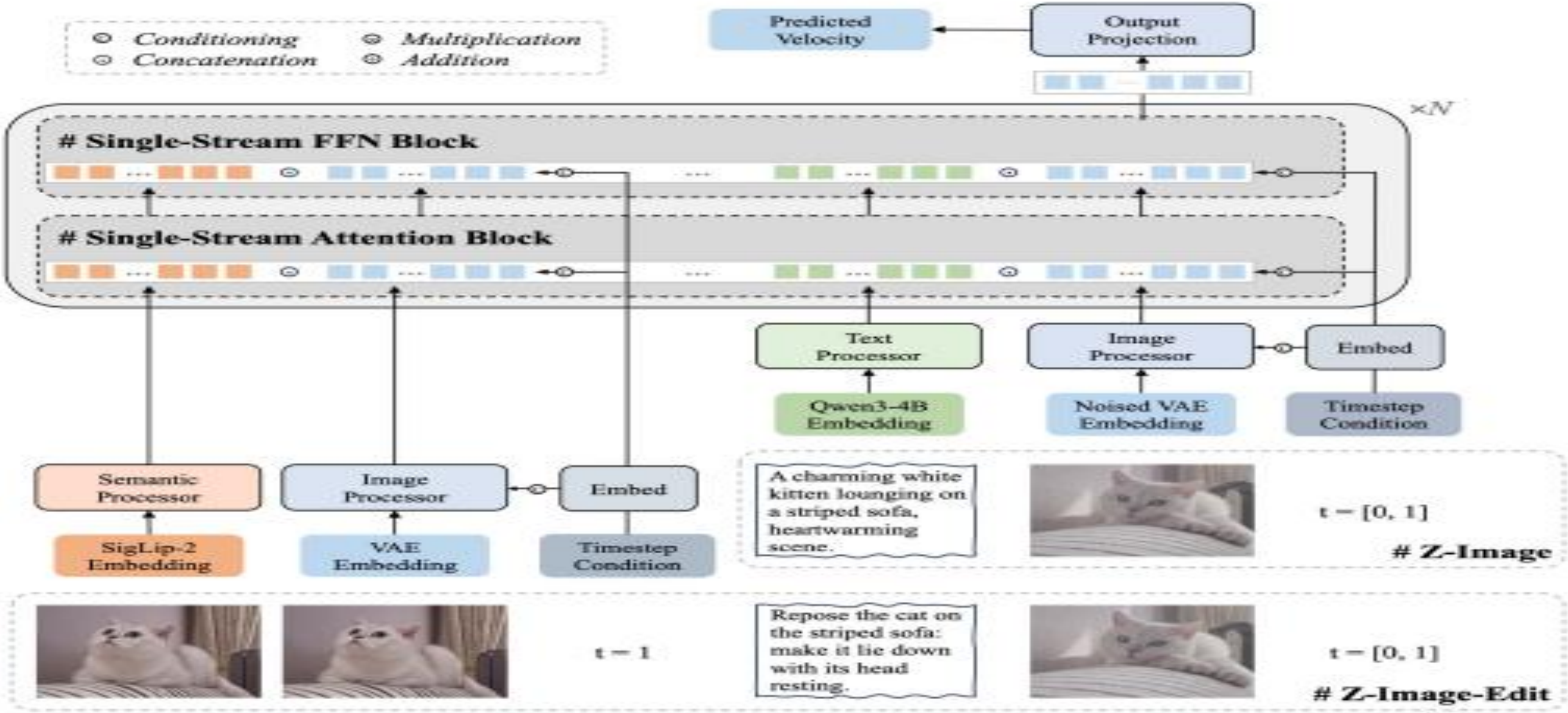
"Double-stream" DiT variant: SD3, Qwen-Image



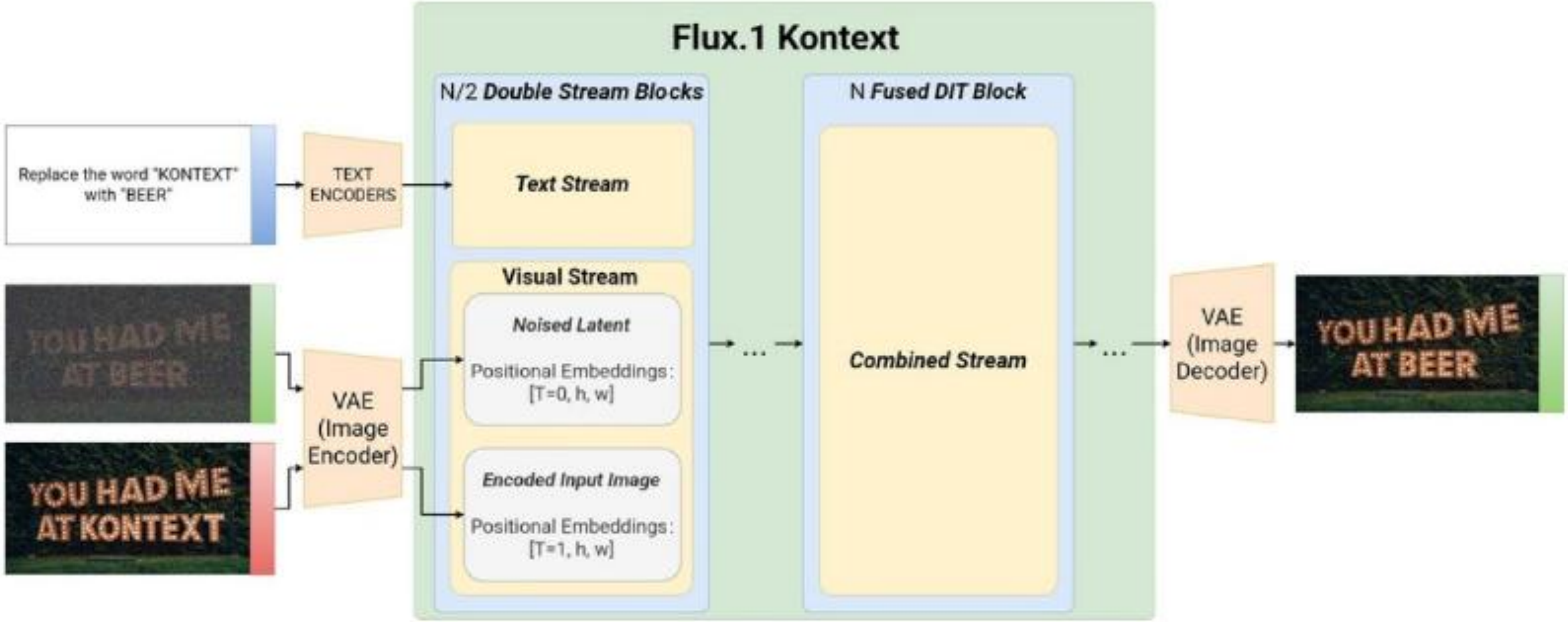
Source Credits: https://cme296.stanford.edu/

"Single-stream" DiT variant: Z-Image



Source Credits: https://cme296.stanford.edu/

Hybrid approach: FLUX.1 Kontext



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Training lifecycle of Multimodal LLMs

Training lifecycle

1	2	3	4
Pre-training	**Post-training**	**Tuning**	**Distillation**
Learn **how** to generate images	Learn how to generate **good** images	Learn how to generate images **for a special case**	Learn how to generate images, **fast**

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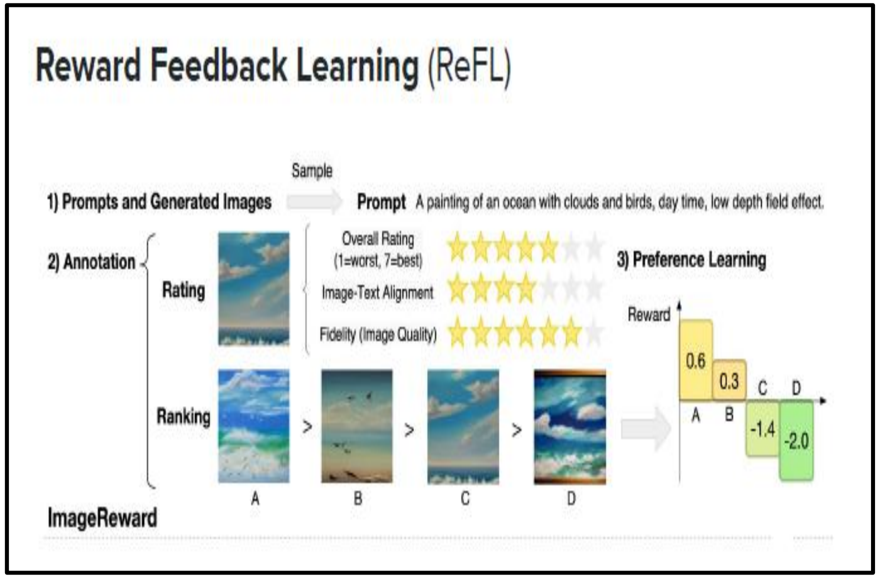
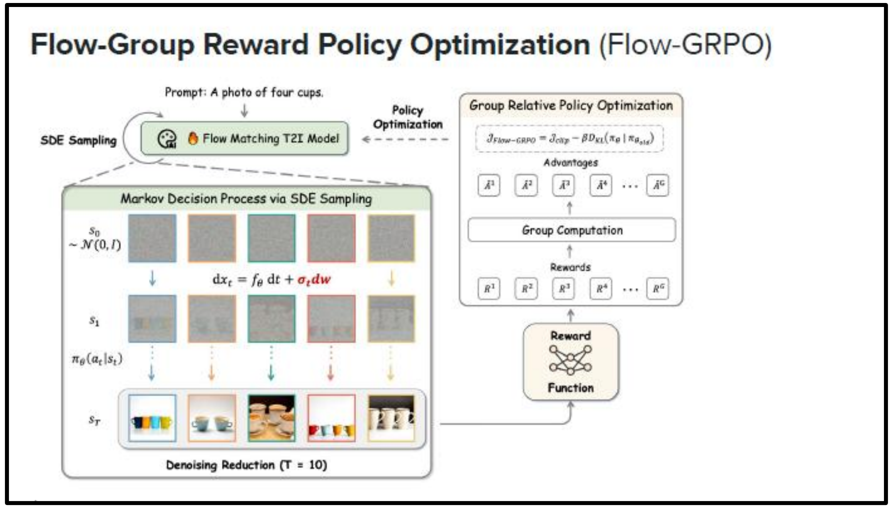
Preference tuning methods





Diffusion-Direct Preference Optimization (Diffusion-DPO)

$$\mathrm{Diff}_{\text{policy}} = \left(\|v_\theta(x_t^{win}, h, t) - v_t^{win}\|_2^2 - \|v_\theta(x_t^{lose}, h, t) - v_t^{lose}\|_2^2 \right)$$

$$\mathrm{Diff}_{\text{ref}} = \left(\|v_{\text{ref}}(x_t^{win}, h, t) - v_t^{win}\|_2^2 - \|v_{\text{ref}}(x_t^{lose}, h, t) - v_t^{lose}\|_2^2 \right)$$

$$\mathcal{L}_{DPO} = -\mathbb{E}_{h,(x_0^{win}, x_0^{lose}) \sim \mathcal{D}, t \sim \mathcal{U}(0,1)} \left[\log \sigma \left(-\beta \left(\mathrm{Diff}_{\text{policy}} - \mathrm{Diff}_{\text{ref}} \right) \right) \right]$$

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Optimizing input prompt

PE = **P**rompt **E**nhancement

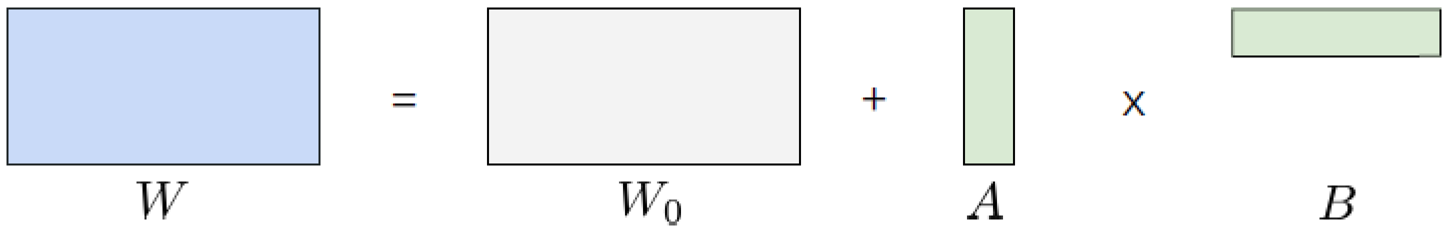

```
A warm, intimate indoor reading scene featuring a plush teddy bear as the central subject.
The teddy bear is seated comfortably in a soft, textured armchair, positioned slightly angled
toward the viewer. It has light beige fur with realistic, finely detailed fibers, giving it a
soft and touchable appearance. The bear wears a cozy, knitted brown sweater with visible wool
texture and subtle stitching imperfections for realism. Perched on its face are small, round,
thin-wire eyeglasses, slightly oversized, resting naturally on its snout. The teddy bear is
holding an open hardcover book titled "The Tales of Woodland", with an elegant, muted green
cover and gold embossed serif typography. The book appears slightly worn but well cared for,
with visible page thickness and gentle curvature where it is being held. The bear's gaze is
directed downward toward the book, creating a calm, focused, and contemplative mood. Lighting
is soft, warm, and cinematic, coming primarily from a table lamp positioned to the left of
the frame. The lamp emits a golden, diffused glow, casting gentle shadows and creating a
cozy, evening ambiance. The lighting should produce a shallow depth-of-field effect, with the
subject sharply in focus while the background softly blurs (bokeh effect). In the foreground,
a ceramic mug sits on a wooden side table. The mug is off-white with a subtle illustration of
a bear printed on it, reinforcing the theme. The mug contains a warm beverage, with faint
steam rising, suggesting warmth and comfort. The wooden table has a natural grain texture,
slightly worn edges, and a matte finish…[truncated]
```

Fine Tuning: only changing a small number of weights

LoRA = **Lo**w **R**ank **A**daptation

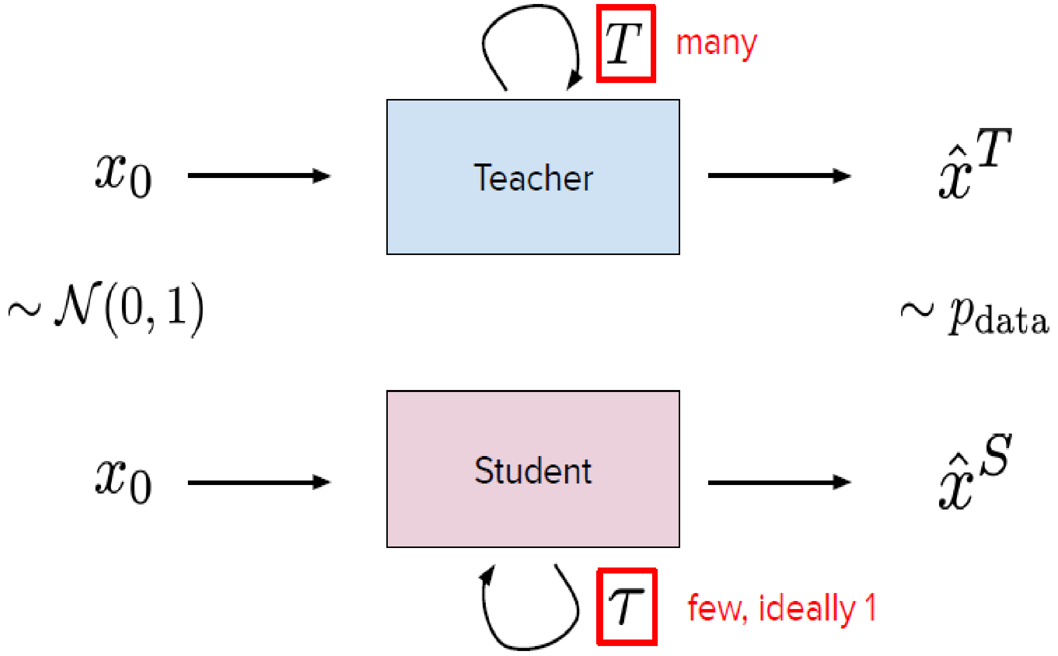
Context. Tuning is resource intensive and not everyone has big GPUs.

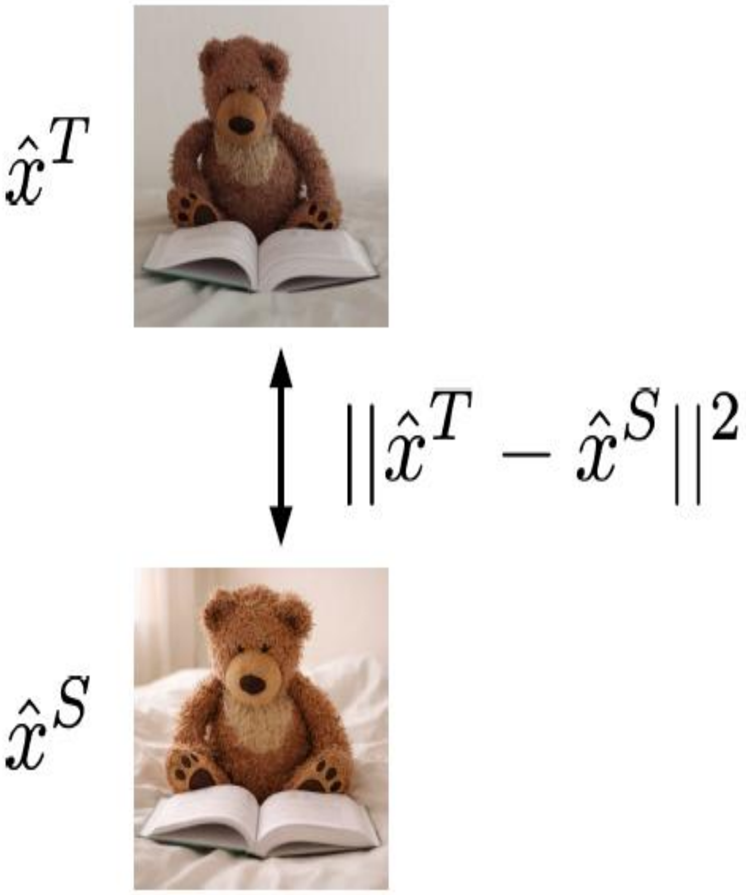
Idea. Low-Rank Adaptation (LoRA) approximates matrix with product of two low-rank matrices



Discussion. Fraction of parameters need to be trained with similar performance

Distillation : Image generation setup





VLM Evaluation

Evaluation

Aesthetics	
"Is this a good picture?"	
• Physical plausibility	• Cleanliness
• Perceptual quality	• Realism

Prompt adherence	
"Did it follow instructions?"	
• Object recall	• Counting
• Text rendering	• Style adherence

A teddy bear reading a book

Other notable dimensions.

- 🛡️ Safety
- 🧩 Diversity
- 🧠 Memorization
- 🔍 Bias

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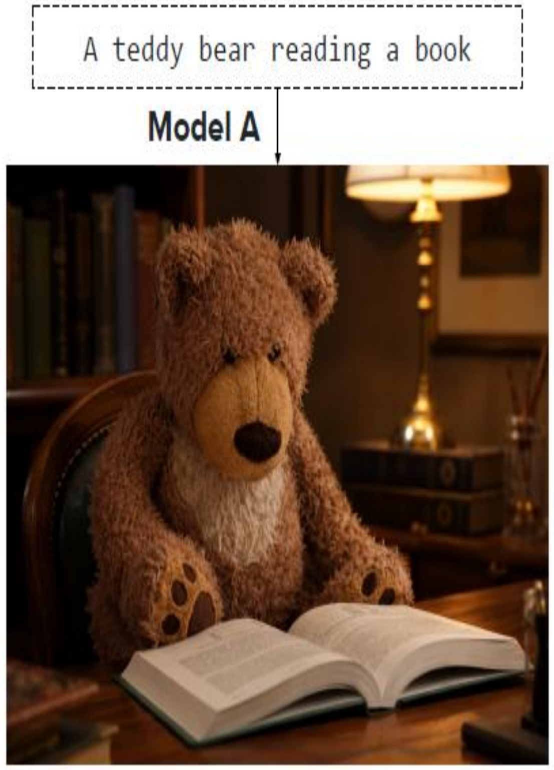
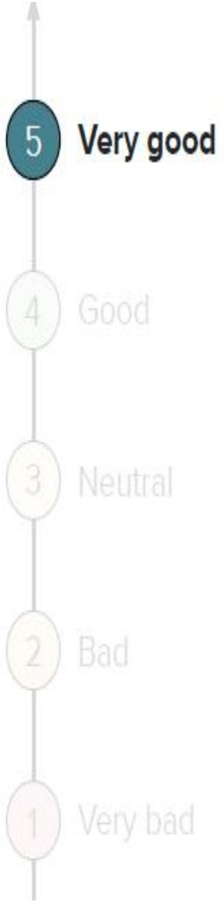
Humans rate images on a scale





$$\text{Score} = \underbrace{\frac{\text{Sum of ratings}}{\text{Number of ratings}}}_{\text{Average of ratings}}$$

Discussion.

- **Good** to have **nuance** in score
- Ratings may be **noisy**
- **Hard** to rate on an **absolute** scale

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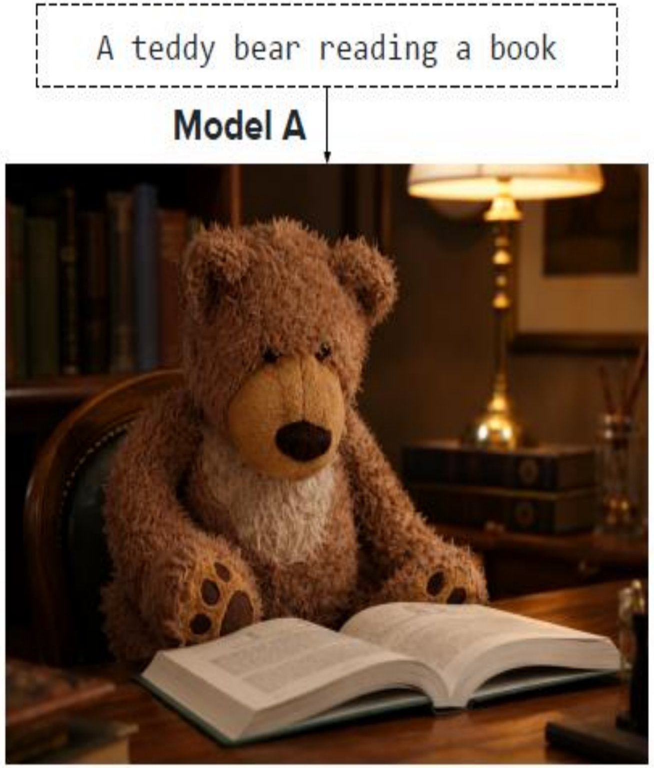
Rate images on a **binary** scale



1 Good

or

0 Bad

Score = $\dfrac{\text{Sum of ratings}}{\text{Number of ratings}}$

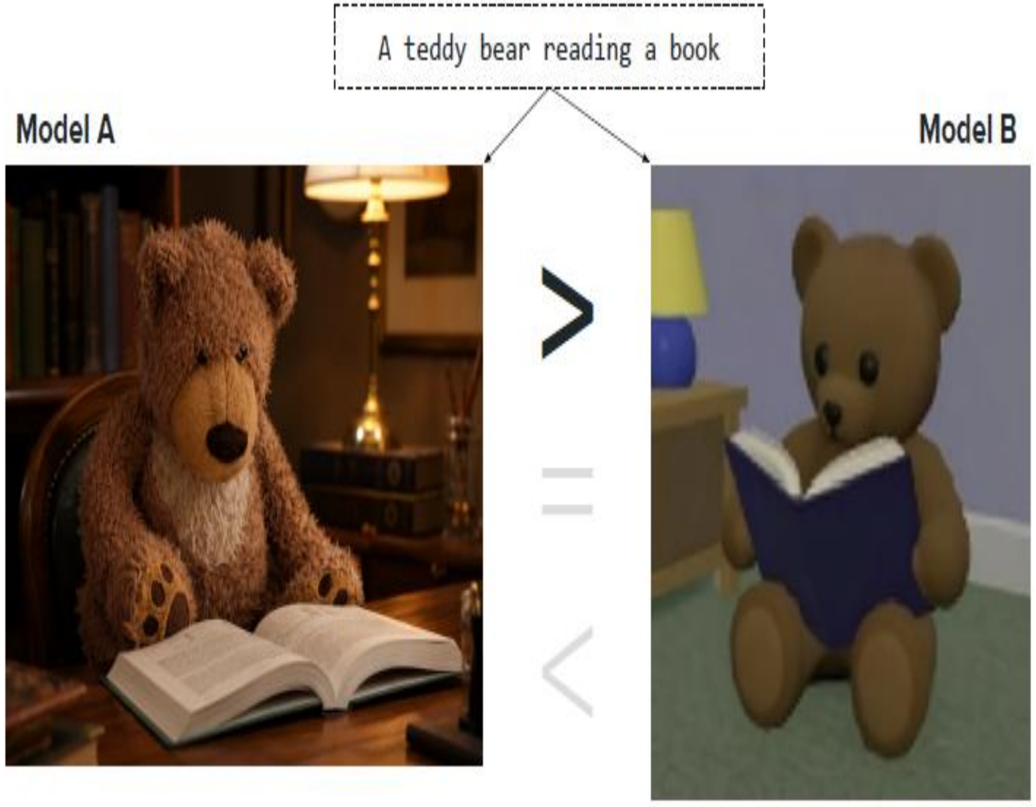
Proportion of pass

Discussion.

- **Easier** task compared to nuanced scale
- Still **hard** to rate on an **absolute** scale

Source Credits: https://cme296.stanford.edu/

Rate images via pairwise comparisons



$$\text{Score} = \boxed{\frac{\text{Number of wins}}{\text{Number of comparisons}}}$$

Win rate

Discussion.

- **Much easier** task than absolute scale!
- Win rate **depends on** who we are comparing with who...

Distance between real and generated distributions

Idea. Quantify distance between generated and real images

$$\underbrace{\|\mu_r - \mu_g\|^2}_{\textbf{Location}\text{ difference}} + \underbrace{\mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right)}_{\textbf{Shape}\text{ difference}}$$

Fréchet inception distance

FID = **F**réchet **I**nception **D**istance

Idea. Quantify distance between generated and real images

$$\mathrm{FID} = \underbrace{\|\mu_r - \mu_g\|^2}_{\textbf{Location}\text{ difference}} + \underbrace{\mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\right)}_{\textbf{Shape}\text{ difference}}$$

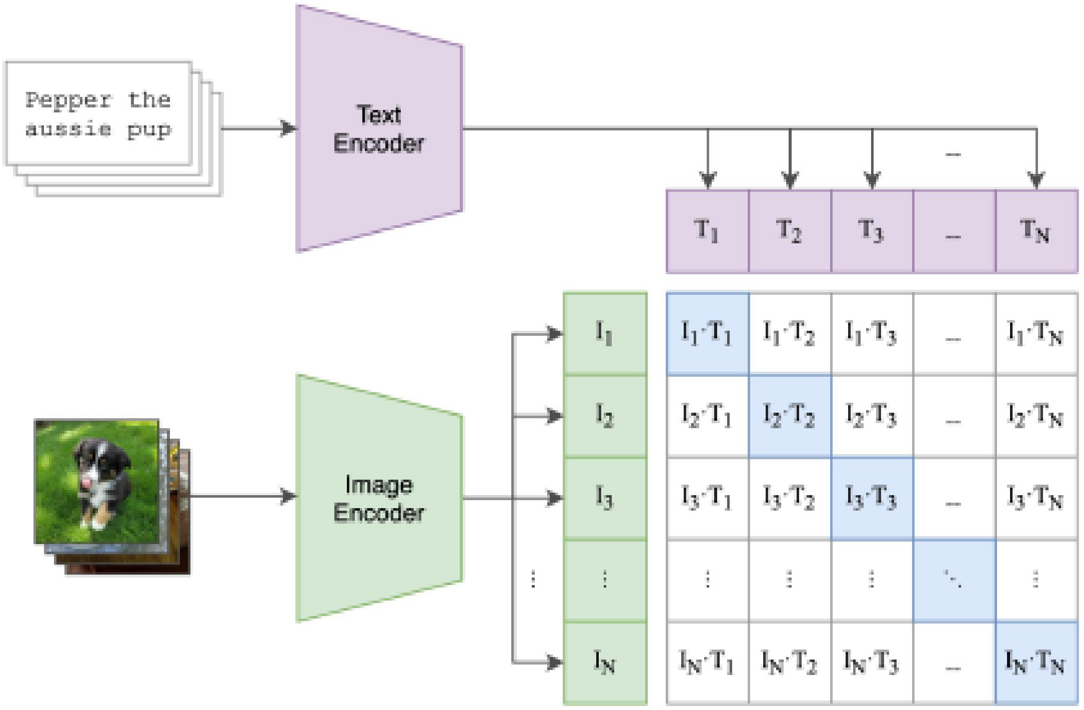
The **lower**, the **better!**

What it measures.

- **Location difference**. Quality, style
- **Shape difference**. Diversity, variety

Prompt Adherence Score

CLIP = Contrastive Language-Image Pretraining



Pixel-wise comparisons

MSE = **M**ean **S**quared **E**rror

Idea. Pixel-wise distance

$$\mathrm{MSE}(x, \widehat{x}) = \frac{1}{HWC} \sum_{i=1}^{H} \sum_{j=1}^{W} \sum_{c=1}^{C} (x_{ijc} - \widehat{x}_{ijc})^2$$

Discussion. Not interpretable and sensitive to pixel position

Pixel-wise comparisons

PSNR = **P**eak **S**ignal-to-**N**oise **R**atio

Idea. Normalized pixel-wise distance

$$\mathrm{PSNR}(x, \widehat{x}) = 10 \log_{10} \left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}(x, \widehat{x})} \right)$$

Discussion. More interpretable but still sensitive to pixel position

Structure comparison

- **Luminance**. Same brightness? ⟶ **Mean** μ
- **Contrast**. Same variance? ⟶ **Variance** σ^2
- **Structure**. Correlation between pixels? ⟶ **Covariance** $\sigma^2_{x\hat{x}}$

- **Luminance** similarity $\dfrac{2\mu_{x_j}\mu_{\widehat{x}_j} + c_1}{\mu^2_{x_j} + \mu^2_{\widehat{x}_j} + c_1}$
- **Contrast** similarity $\dfrac{2\sigma_{x_j}\sigma_{\widehat{x}_j} + c_2}{\sigma^2_{x_j} + \sigma^2_{\widehat{x}_j} + c_2}$
- **Structure** similarity $\dfrac{\sigma_{x_j\widehat{x}_j} + c_3}{\sigma_{x_j}\sigma_{\widehat{x}_j} + c_3}$

$$\mathrm{SSIM}(x_j, \widehat{x}_j) = \underbrace{[l(x_j, \widehat{x}_j)]}_{\textbf{Luminance}\text{ similarity}} \cdot \overbrace{[c(x_j, \widehat{x}_j)]}^{\textbf{Contrast}\text{ similarity}} \cdot \underbrace{[s(x_j, \widehat{x}_j)]}_{\textbf{Structure}\text{ similarity}}$$

$$\mathrm{SSIM}(x_j, \widehat{x_j}) = \frac{(2\mu_{x_j}\mu_{\widehat{x_j}} + c_1)(2\sigma_{x_j\widehat{x_j}} + c_2)}{(\mu^2_{x_j} + \mu^2_{\widehat{x_j}} + c_1)(\sigma^2_{x_j} + \sigma^2_{\widehat{x_j}} + c_2)}$$

Feature comparison

LPIPS = **L**earned **P**erceptual **I**mage **P**atch **S**imilarity

Idea. Use pre-trained model to **compute features** that align with **visual perception**

$$\mathrm{LPIPS}(x, \widehat{x}) = \sum_{l} \frac{1}{H_l W_l} \left\| w^l \odot (\phi_l(x) - \phi_l(\hat{x})) \right\|^2$$

Limitations. Not directly interpretable

Common benchmarks for evaluation of multimodal LLMs

Object alignment

- Verify objects, counts, colors
- Tests compositional image generation
- Fine-grained failure diagnosis

GenEval

Dense prompts

- Follow long detailed prompts
- Track entities, attributes, relations
- Evaluates semantic completeness

DPG-Bench

Text rendering

- Generate readable text in images
- Handles long strings
- Tests OCR-level fidelity

LongText-Bench

Image editing

- Apply user edit instructions
- Preserve source image quality
- Covers real-world edit types

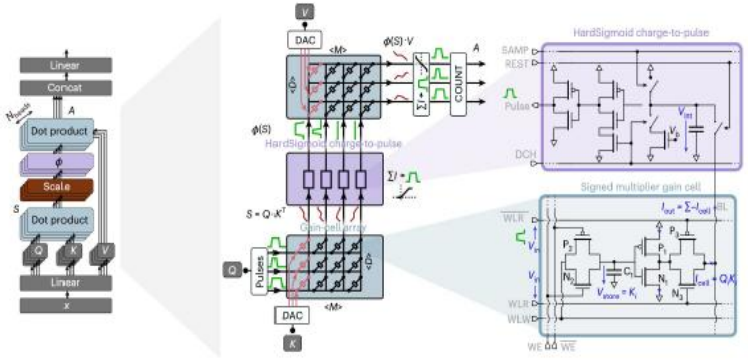
GEdit-Bench

Trends in Generative Image Models

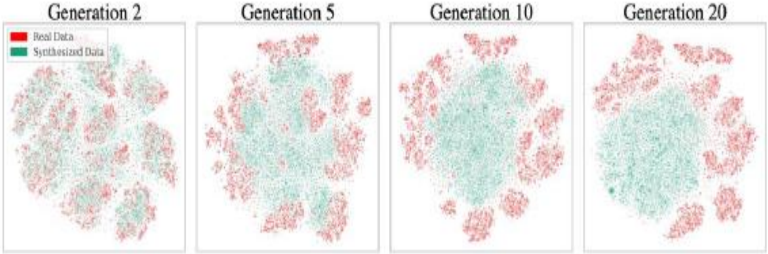
- **Architecture**. Decoder with joint attention.
- **Capabilities**.
  - Work across resolutions
  - Spatial awareness
  - OCR
  - Includes other modalities
  - Reasoning

Ongoing challenges

- Costs

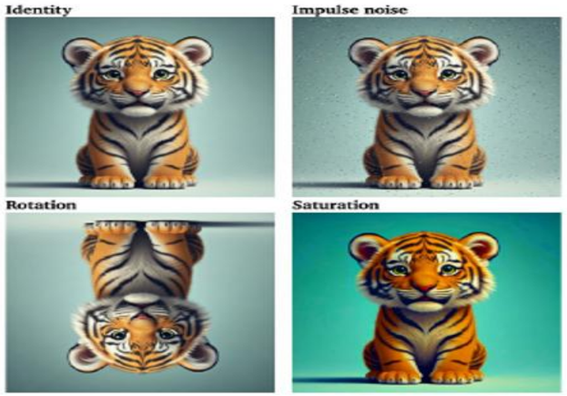


- Data quality



"Model collapse"

- Trust



Watermarking

- Safety

TECHNOLOGY

Deepfakes laws not moving at the speed of AI

Resources

- https://visionbook.mit.edu/stat_image_models_revised.html
- Stanford Course CME296, https://cme296.stanford.edu
- Image Processing Fundamentals, https://cmp.felk.cvut.cz/cmp/courses/dzo/resources/course_ip_tudelft/course/fip-Statisti.html
- Probabilistic and Semantic Descriptions of Image Manifolds and Their Applications, https://arxiv.org/pdf/2307.02881